# The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock on the $W(3,3)$ Substrate

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**Abstract**

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3,3)$: forty points, forty lines, automorphism group with faithful projective subgroup $\mathrm{PSp}(4,\mathbb{F}_3)$ of order 25920 and full Weyl extension $W(E_6) \cong \mathrm{PSp}(4,3){:}2$ of order 51840. The symplectic double cover $\mathrm{Sp}(4,3)$ also has order 51840, but its central involution is invisible on projective objects; the two extensions must not be conflated. The photon carries $W(3,3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine's states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the dual-torsor compiler (three regular copies of both the flat $\mathbb{F}_3^3$ program shell and the noncommutative Heisenberg state shell, with canonical 27+27+27 triality selected by the Heisenberg center), the oriented control plane (top homology $H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30$, whose conjugate 5-sectors fuse to one Weyl-visible 10-channel), the mirror middleware (a 2160-slot $D_{12}$ transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget 36/40 and contextual fraction 1/10), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon's own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). We also give the recursive engineering law: a level-$n$ holonet has $40^n$ photonic leaves, $(40^n - 1)/39$ $W(3,3)$ instances, and reversible routing diameter bounded by $8n = 8\log_{40} N$, while durable commits use a separate Császár/tomotope clock $T(n) = 4(7^n - 1)$. The runtime contains an exact parabolic packet decoder: the Bell-line stabilizer splits the 1296 compass incidences into 648+324+162+162, giving the complete binary prefix code $\{0, 10, 110, 111\}$ for mirror, schedule, and two chiral caches, while the dual point stabilizer supplies two persistent 648-slot address sheets. The two cache branches are now resolved internally: each is the same homogeneous space $H/C_4$, each is a two-sheet cover of one canonical 81-route base $H/D_8$, and their 324-slot union has full equivariant commutant $D_8$ acting in 81 four-state fibers. This split address bus is provably distinct from the older non-split ternary $81 \to 162 \to 81$ transport memory. At the communication edge, the Witting delayed-query desk is now a frame compiler: 40 tetrads times 4 Alice slots times 4 Bob slots give a 640-entry admission ROM, with 480 off-diagonal data handshakes and 160 diagonal contextual witness apertures. The corresponding analyzer bank is sparse physical optics rather than a generic multiport: the 40 Witting bases split as 1+12+27 analyzers with at most 12 nonzero matrix entries and only cube-root phase weights over $\sqrt{3}$. The newest storage layer turns the tomotope cover pathology into an engineering feature: durable storage is cover-indexed by a $\mathbb{Z}_k^3$ fiber over the stable 48-block packet ABI, sentinel-aware

compilation can trade commit-time slack for lower $g = 15$ fault energy, the first exact guard band is the arithmetically forced desynchronization at $n = 5$ with remainder $24 = f$, and the Grünbaum–Coxeter Wythoff operations of the tomotope are exactly the packet-growth ladder $4, 12, 24, 48, 96$. No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

## Contents

# 1 Overture: the machine is the geometry is the carrier

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

**Principle 1.1** (Identity of the three layers)**.** *Transporting the photon through the mesh* is *applying a gate* is *routing a packet. One physical process; three descriptions.*

The identification rests on seven exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§2.1): the same $W(3,3)$ is drawn by the photon's *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).
2. **Routing–gate identity** (§4): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4, \mathbb{F}_3)$.
3. **Parabolic address–route duality** (§2.3): the point parabolic supplies two 648-slot address sheets, whereas the Bell-line parabolic decodes the compass fabric by the complete prefix code $0, 10, 110, 111$; its cache quarter is an exact $D_8$-over-81 four-state address fiber. Address and route are the two nodes of the same $C_2$ building, while the separate ternary transport extension carries temporal state.
4. **Dual-torsor state–program compilation**: the Steinberg memory restricts to both canonical order-27 shell groups as three regular copies. Bell-shell programs use commuting $\mathbb{F}_3^3$ coordinates; point-shell states use Heisenberg $3^{1+2}$ coordinates; both compile into the same 81 logical qutrits.
5. **Mirror middleware** (§5): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the $C_{12}$ side but routed by the $D_{12}$ side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.
6. **Universality from optics** (§6.5): the photon's own beam-splitting elements generate the complete Clifford group (symplectic closure $= 51840$, exact), and its matter shell is precisely its magic supply.
7. **Fractal holonet scaling** (§7): recursively replacing every site by a fresh $W(3,3)$ core gives a universal computer/network with exact $40^n$ leaves and logarithmic reversible routing.

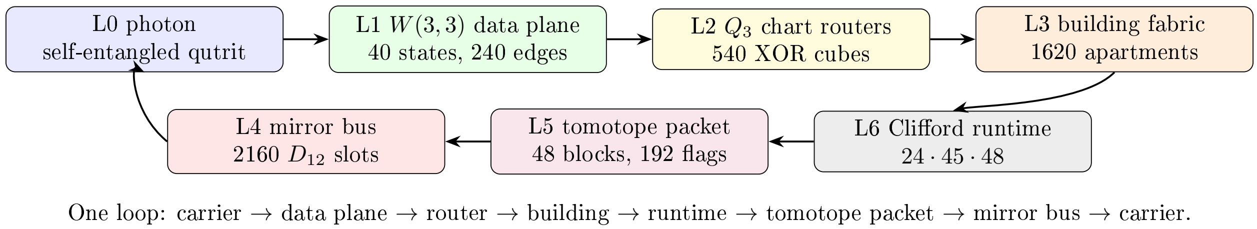


Figure 1: The photonic holonet as an engineering stack. The same photon is data, instruction, and packet; the same finite geometry supplies the network, middleware, and runtime.

# 2 The substrate

**Definition 2.1** ($W(3,3)$)**.** *Let $V = \mathbb{F}_3^4$ with the alternating form $\langle u, v\rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. The points of $W(3,3)$ are the 40 projective points of $V$; the lines are the 40 totally isotropic projective lines. Each point lies on 4 lines and each line carries 4 points; collinearity defines the strongly regular graph* $\mathrm{SRG}(40, 12, 2, 4)$.

Three related symmetry groups occur, and their roles are different:

$$\underbrace{\mathrm{Sp}(4,3)}_{2\cdot\mathrm{PSp}(4,3),\ \text{Clifford double cover}}, \qquad \underbrace{\mathrm{PSp}(4,3)}_{25920,\ \text{faithful projective action}}, \qquad \underbrace{W(E_6) \cong \mathrm{PSp}(4,3){:}2}_{51840,\ \text{outer Weyl extension}}.$$

The first and third groups both have order 51840 but are not the same extension: the center of $\mathrm{Sp}(4,3)$ acts trivially on projective points and lines, while the outer involution in $W(E_6)$ acts nontrivially. The projective group acts with permutation rank 3. The ATLAS index-40 subgroups are the two building parabolics $3^{1+2}{:}2A_4$ and $3^3{:}S_4$ [18]. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

**Formal grounding (the master formalism).** In the master manuscript [1] the whole structure is forced by one Diophantine seed: the *master equation* $q! = 2q$ has the unique positive-integer solution $q = 3$, and the SRG quadratic $x^2 - q!\,x + 2^q = (x-2)(x-4)$ then delivers $\lambda = 2$, $\mu = 4$ and the full parameter closure ($k = 2^q + q + 1$, $v = 40$, multiplicities $f = 24$, $g = 15$, cyclotomic ladder $\Phi_3 = 13$, $\Phi_6 = 7$, $\Phi_{12} = q^4 - q^2 + 1 = 73$, $\Theta = q^2 + 1 = 10$). Two of those closure constants reappear in this paper as *spectral field discriminants*: the chart-web's irrational eigenvalue pairs $1 \pm \sqrt{10}$ and $(-1 \pm \sqrt{73})/2$ (§4) live precisely in $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ — the machine's transport spectrum speaks the same cyclotomic dialect as the physics tier ladder.

## 2.1 The two carriers and the operator–operand duality

**Theorem 2.2** (Two realizations, one geometry)**.** (i) *The 40 Witting rays in $\mathbb{C}^4$ (the photon's path$\otimes$polarization space) have orthogonality graph isomorphic to the collinearity graph of $W(3,3)$.* (ii) *The 40 projective classes of two-qutrit Pauli displacements $D(v) = X^aZ^b \otimes X^cZ^d$ on $\mathbb{C}^9$ (the photon's past$\otimes$future time-bin space) have commutation graph equal to the same $W(3,3)$, via $D(u)D(v) = \omega^{\langle u,v\rangle}D(v)D(u)$.* Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40\times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

## 2.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\mathrm{PSp}(4,3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

| index | subgroup | points | lines | vacuum | cubic surface |
|---|---|---|---|---|---|
| 27 | $2^4{:}A_5$ | $[40]$ | $[40]$ | homogeneous (icosahedral) | 27 lines |
| 36 | $S_6$ | $[40]$ | $[10,30]$ | spread fibration | 36 double-sixes |
| 40 | parabolic | $[1,12,27]$ | $[4,36]$ | holonet point vacuum | trihedra I |
| 40 | parabolic | $[4,36]$ | $[1,12,27]$ | dual line vacuum | trihedra II |
| 45 | $(2T\times 2T){:}2$ | $[8,32]$ | $[16,24]$ | binary polar | 45 tritangents |

The physical decomposition $40 = 1+12+27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group $O_h$ ($= S_4 \times \mathbb{Z}_2$, order 48) is the chart symmetry of §4, and the binary icosahedral group $2I = \mathrm{SL}(2,5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits $[20,20]$ and line orbits $[10,30]$ echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808-bt813`.

## 2.3 The parabolic router: address and route are dual

The two index-40 parabolics are not duplicate descriptions of one local neighborhood. They execute different halves of a packet protocol. This becomes visible only after the compass census: there are two 216-element classes of icosahedral $A_5$ cores, and exactly $1296 = 216 \cdot 6$ cross-class pairs meet in $D_{10}$.

**Theorem 2.3** (Bell–compass parabolic router)**.** *Fix an isotropic Bell line. Its projective Siegel parabolic $3^3{:}S_4$ has four orbits on the 1296 incident compass pairs:*

$$162_L + 162_R + 324 + 648.$$

*They are distinguished objectwise by the Bell line lying in the pentad core's $5_L$, $5_R$, 10, or 20 line orbit. The normalized orbit sizes satisfy the Kraft equality*

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} = 1,$$

*so they admit the complete prefix decoder*

$$110 \mapsto \textit{left cache}, \quad 111 \mapsto \textit{right cache}, \quad 10 \mapsto \textit{schedule}, \quad 0 \mapsto \textit{dark mirror}.$$

*The outer Weyl involution fuses exactly the two 162 cache orbits, leaving $324_{\text{cache}} + 324_{\text{schedule}} + 648_{\text{mirror}}$. By contrast, the dual point parabolic has two regular projective orbits of size 648; the outer extension preserves rather than fuses these address sheets.* Witness: `bt859_bell_compass_parabolic_router.py`.

The equality case of Kraft–McMillan means the routing words fill a binary decision tree with no unused branch [19]. Here the code lengths were not optimized from assumed traffic probabilities: they were read from exact group orbits. A packet first asks whether its Bell line is in the dark mirror half; if not, whether it is in the common schedule quarter; only the remaining cache quarter needs the third, chiral bit. The decoder is self-delimiting and constant-depth, with expected word length

$$\frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}$$

under the invariant uniform measure on the 1296 compass pairs.

| orbit | Bell-line position | prefix | engineering action |
|---|---|---|---|
| 648 | dark 20-orbit | 0 | enter mirror bus |
| 324 | common schedule 10-orbit | 10 | timetable-local route |
| 162 | absolute pentad $5_L$ | 110 | left cache |
| 162 | absolute pentad $5_R$ | 111 | right cache |

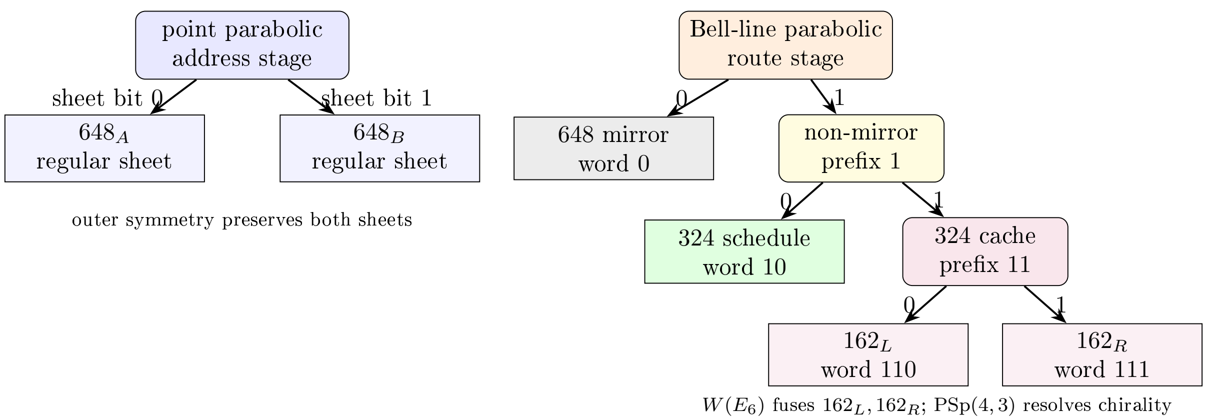


Figure 2: The two $C_2$ parabolics implement a dual protocol. Point localization selects one of two address torsors; Bell-line localization decodes a complete variable-length route word.

This theorem also closes an honesty boundary. The Bell stabilizer and the compass incidence set both have cardinality 1296, but they are not a regular $G$-set identification. Projectively, the Bell stabilizer has order 648 and the orbit stabilizers are $1, 2, 4$. The useful result is stronger than the count: a hierarchical decoder and a persistent address bit are forced by the two parabolic actions.

# 3 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon's own registers.

## 3.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \mapsto \frac{1}{\sqrt{2}}(|H, a\rangle + |V, b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 2.2(i).

### 3.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate $F_3$; a three-bin delay ladder $(0, \tau, 2\tau)$ defines past and future registers; one electro-optic modulator applies $\mathrm{CX}_{p\to f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle \;=\; \mathrm{CX}_{p\to f}\,(F_3 \otimes I)|0\rangle_p|0\rangle_f \;=\; \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p|j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any $U$ in the future arm, the recombination visibility is the *trace–Choi witness*

$$V(U) \;=\; \frac{|\mathrm{Tr}\,U|}{3}, \qquad V(F_3) = \tfrac{1}{3}, \quad V(X) = V(Z) = 0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness:* `bt820_self_entanglement_protocol.py`.

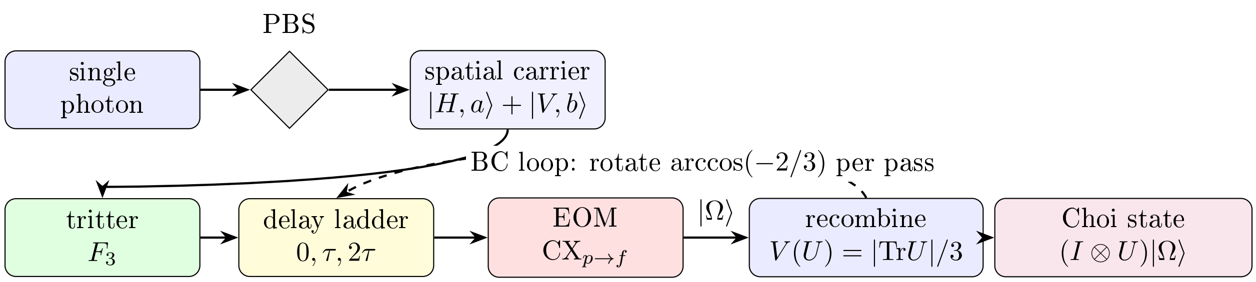


Figure 3: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §11.

# 4 The network: atlas, building, and routing

**Theorem 4.1** (The hypercube atlas)**.** $W(3,3)$ *contains exactly* 540 *skew line pairs. For each pair the non-collinearity graph on its* 8 *points is the cube graph* $Q_3$*, with full symmetry group* $O_h$ *of order* 48 *(these are the only order-*48 *stabilizers; the binary* 2$O$ *does not occur). Each chart admits an exact* $\mathbb{F}_2^3$ *Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every* $W(3,3)$ *nonedge is a chart edge in exactly* $12 = k$ *charts; every edge is an antipode pair in exactly* $9 = q^2$. Witnesses: `bt773`, `bt777`, `bt778`.

### 4.1 Hypercube networking, not hypercube decoration

The cube is not only a picture for one chart. It is the local interconnect primitive. A chart gives eight simultaneously valid addresses

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111 \in \mathbb{F}_2^3,$$

and the elementary network instruction is the dimension-order hypercube hop

$$a \longmapsto a \oplus e_i, \qquad i \in \{1,2,3\}.$$

This is the standard e-cube rule of hypercube networks: compare source and target, flip one differing coordinate at a time, and finish in at most three local hops [10]. In the holonet that same XOR flip is also a register operation, because the chart address is a finite "which-register" word on the photon's self-entangled carrier. The local routing primitive is therefore simultaneously

$$\text{bit flip} \quad = \quad \text{qutrit-frame update} \quad = \quad \text{optical path choice.}$$

A packet in the photonic holonet is not merely a payload. The natural packet header is

$$\mathcal{P} = (\text{payload},\ \text{Pauli frame},\ \text{chart word},\ \text{apartment hop},\ \text{mirror slot},\ \text{clock phase}).$$

The payload is the quantum state or teleported gate; the Pauli frame is the two-trit feedforward record; the chart word is the $\mathbb{F}_2^3$ route inside the current cube; the apartment hop selects the next chart; the mirror slot chooses the $D_{12}$ transport bus of §5; and the clock phase is the cyclic $C_{12}$ interrupt that keeps the selector coherent. This is why the same object can be called a computer architecture and a network architecture. A node never sends data to a processor: the act of transport is the processor's gate action.

**Theorem 4.2** (The fabric is the building)**.** *The chart-adjacency web (*540 *nodes, two charts adjacent when one's axis is a disjoint transversal pair of the other) is* 6*-regular with exactly* 1620 *edges, and these edges are in canonical bijection with the* 1620 *apartments of the Tits building of* $\mathrm{Sp}(4, \mathbb{F}_3)$*. The web has diameter* 5*; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a* 15*-dimensional sentinel eigenspace.* Witnesses: `bt777`, `bt779`, `bt744`.

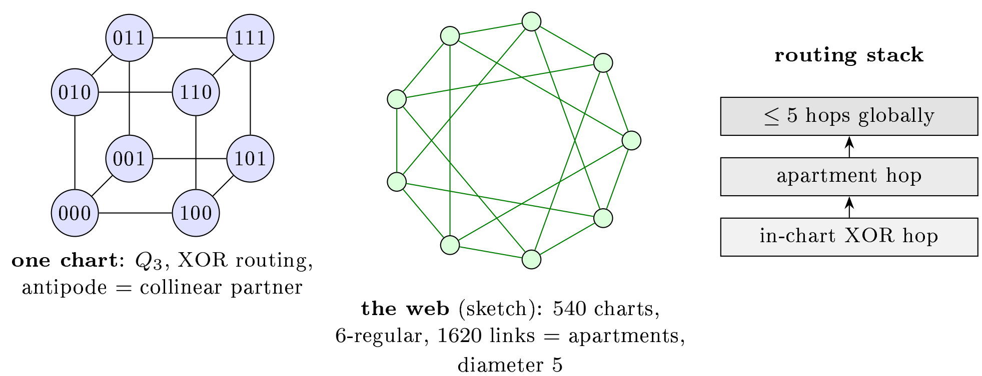


Figure 4: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§6), routing is computation. At scale the recursion is the holonet ladder: $40^n$ cores at level $n$, the same $\mathrm{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

**Corollary 4.3** (Two-plane routing bound)**.** *Inside a chart the worst route length is* 3*, the diameter of* $Q_3$*. Between charts the worst route length is* 5*, the diameter of the apartment web. One recursive digit of a holonet address therefore costs at most*

$$3 + 5 = 8$$

*reversible transport moves. BT827 globalizes this to* $8n = 8 \log_{40} N$ *for* $N = 40^n$ *photonic leaves.*

# 5 The middleware: tomotope mirrors and the runtime bus

**Definition 5.1** (Tomotope packet)**.** *In this paper the tomotope is used in the only way the finite calculation currently justifies: as a local middle-layer incidence packet. A tomotope packet has four vertex carriers, twelve local edge axes, sixteen triangular face labels, and forty-eight edge–face incidence blocks. The doubled base/shadow fiber gives* 192 *flags. Thus the tomotope is the durable packet format sitting between the fast cube router and the global mirror bus; it is not introduced as an ambient continuous torus, and it is not identified with the rectangle clock.*

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is $C_{12}$ and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is $D_{12}$ and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

**Theorem 5.2** (The $D_{12}$ mirror bus)**.** *The* 540 *cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has* $540 \cdot 4 = 2160$ *slots. The map*

$$(\textit{chart},\ \textit{common transversal line}) \longmapsto (\textit{chart},\ \textit{antipode pair cut out by that transversal})$$

*is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order* 12 *and GAP identifies it as* $D_{12}$*, not* $C_{12}$*.* Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the tomotope middle layer. Each transversal is a $K_4$, and each $K_4$ has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \qquad 4 \cdot 3 = 12 \text{ edge labels}, \qquad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

edge–face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3 \rangle$ tomotope block layer, not an analogy to a toroidal polyhedron. *Witness:* ***bt814_ tomotope_ middle_ layer_ from_ residual_ tetrahedra. py.***

**Theorem 5.3** (Photonic mirror middleware)**.** *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

*Here* $24 = f$ *is the full* $\mathrm{Sp}(4,3)$ *lift of the projective* $D_{12}$ *mirror-slot stabilizer,* 45 *is the hyperbolic polar-pair/tritangent geography, and* 48 *is the local tomotope edge–face middle layer (also the order of the chart group* $O_h$*). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomotope middle carrier.* Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomotope local block. In software terms, $C_{12}$ is the clock interrupt, $D_{12}$ is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.

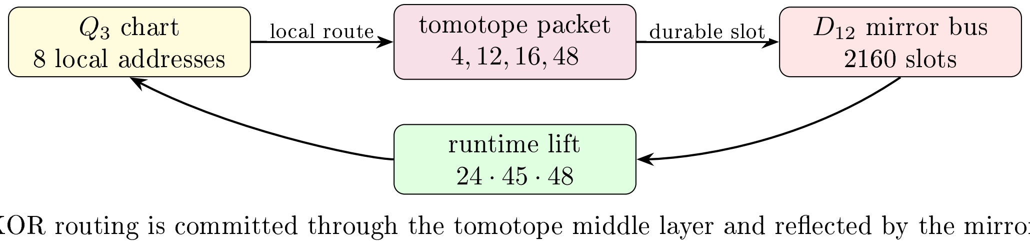


Figure 5: Middleware schematic. The cube is the reversible local router; the tomotope is the edge–face packet format; the $D_{12}$ bus is the global mirror transport object.

### 5.1 Product-grid closure ABI v2

The 12 closure strands are modeled as

$$C_3 \times V_4, \qquad V_4 = C_2 \times C_2,$$

rather than as a bare $C_{12}$. The $C_3$ coordinate is the Szilassi/Fano channel and qutrit phase axis; the $V_4$ coordinate is the four-triangle E6 gauge-shell/D4 branch axis.

The E6 firewall square is now upstream in the claim DAG through the exact nodes

$$36 \to 72, \qquad 72 + 6 = 78, \qquad 72 + 9 = 81.$$

This supports the exact finite ABI/CSS branch while leaving blocked formula and physics claims downstream of the transcription gate.

The closure packet ABI is upgraded to v2. Each packet carries $C_3$ channel, $V_4$ branch, active/guard outputs, 72-row sector metadata, the nine-mode firewall gap, and claim-DAG dependencies.

### 5.2 ABI v2 consumer, E6 claim table v2, and D4/V4 branch audit

The closure ABI v2 is executable. It regenerates 12 packets, 48 events, and 72 rows while preserving the dual-axis metadata. The three $C_3$ channels each carry 24 rows, and the four $V_4$ triangles each carry 18 rows.

The E6-merged claim DAG now produces a table with explicit E6 provenance for the 72-sector, the 81 closure, and the $C_3 \times V_4$ grid. Blocked formula and physical claims remain downstream.

On branch classes, D4 shear fixes each $V_4$ triangle, while $\tau_4$ acts as an order-two translation preserving the triangle partition as a set. The two operations still fail to commute on full branch/phase states, so retwining remains part of the exact decoder rule.

### 5.3 BT1486–BT1488 ABI v2 retwined CSS and splice manifest

BT1486 reruns the CSS join from the ABI v2 consumer output. The result is not just the old 72-row count: every row satisfies the retwined $X$- and $Z$-syndrome checks, while the $C_3$ channel profile remains $P_0 = P_1 = P_2 = 24$ rows and the $V_4$ triangle profile remains $T_0 = T_1 = T_2 = T_3 = 18$ rows.

BT1487 classifies the branch symmetries behind those four triangles. The full triangle-partition stabilizer is $S_4$ of order 24, giving the Fano point reading $7 \cdot 24 = 168$. The physically used square subgroup is $D_4$ of order 8, giving the Fano flag reading $21 \cdot 8 = 168$. Thus the active detector-bin count is Fano-native, not merely an optical count.

BT1488 updates the splice cascade: the E6 claim table v2 from BT1484 supersedes the BT1472 table as the preferred paper table, and the preferred exact finite insert packet is now BT1474, BT1480, BT1482, BT1484, BT1486, and BT1487. The Golden Quartic/Moebius-ball equation fill remains blocked pending rendered formula transcription; no public source-code repository is assumed.

### 5.4 BT1489–BT1491 row symmetry and Fano–E6 square

BT1489 lifts the $S_4$, $D_4$, and $V_4$ branch actions from triangle labels to the actual ABI v2 row layer. All 24 elements of $S_4$, all 8 elements of the square $D_4$ subgroup, and all 4 $V_4$ translations act as honest permutations of the 72 active/guard value rows. The lift preserves the $C_3$ channel, row kind, qutrit value, guard slot, and column formula, so the branch symmetry is now a decoder-row symmetry.

BT1490 is the new finite-geometry lock:

$$24 = 4 \cdot 6 = 3 \cdot 8, \qquad 72 = 3 \cdot 24, \qquad 168 = 7 \cdot 24 = 21 \cdot 8, \qquad 81 = 72 + 9.$$

The shared 24-state fiber is $V_4$ branch times six row-value slots on the ABI side, and three local Fano arms times the $D_4$ flag stabilizer on the Fano side. Thus the same finite fiber feeds the E6/CSS 72-sector, the 81-closure after adjoining the $q^2 = 9$ firewall gap, and the Fano-native 168 active-bin bus.

BT1491 makes the paper splice idempotent. The main holonet paper now imports the BT1480–BT1491 exact finite packet through one controlled block; rerunning the splicer replaces that block rather than duplicating it. The claim firewall remains intact: this is an exact finite ABI and incidence statement, not a detector calibration, optical-layout claim, or imported Golden Quartic/Mobius particle interpretation.

### 5.5 BT1492–BT1494 canonical Fano fiber and pulse release lock

BT1492 removes the last arbitrary index from the 24-state bridge. In the Fano plane, $\mathrm{GL}(3, 2)$ has order 168. Fixing one point leaves a point stabilizer of order 24, and this stabilizer acts as the full $S_4$ on the four Fano lines not through that point. Fixing one incident flag leaves an order-8 flag stabilizer with the $D_4$ order profile. Therefore the shared fiber is canonical:

$$24 = 3 \cdot 8,$$

namely three Fano lines through the anchor point times the $D_4$ flag stabilizer.

BT1493 compiles those row symmetries down to the existing physical holonet interfaces. A row action first selects a BT1411 detector slot, then hands that slot to the BT1374 rule mirror_slot mod 4, and finally occupies the matching BT1407 Hesse epilogue lane for its $C_3$ channel and row-value slot. All 24 $S_4$ actions compile across all 72 rows, while the order-8 $D_4$ square subgroup is the native square-pulse layer.

BT1494 repairs the broad photonic-QEC release lock. The legacy tests expected selected `PART_*` artifacts at the repository root, while the preserved copies lived under `manuscripts/parts`. The repair restores those root release-lock targets and regenerates the live scheduler, harmonic bus, fusion splice, and projective screen/bulk/QEC artifacts.

# 6 The software: braids, teleported gates, universality

## 6.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual ($\pm$) encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`*,* `bt741`*.*

## 6.2 The photon is its own gate

A gate $U$ is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs $U$ (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`*.*

## 6.3 Instruction set architecture

The practical instruction set is finite:

$$\mathcal{I}_{\mathrm{holo}} = \{F_p, F_f,\ S_p, S_f,\ \mathrm{CX}_{p\to f},\ \sigma^5 = Z,\ D_{12}\text{-mirror},\ M_{36}\text{-magic}\}.$$

$F$ and $S$ generate single-qutrit Clifford frame changes; CX couples past to future; $\sigma^5 = Z$ gives exact braid-register flips; the $D_{12}$ mirror operation moves a packet through the atlas without confusing it with the cyclic selector clock; and $M_{36}$ denotes injection from the thirty-six magic rays of §9. The Clifford part normalizes Pauli frames and is efficiently classically trackable; it is not universal by itself. Universality starts only when the Clifford runtime can consume the intrinsic magic sector. This is the engineering meaning of "matter equals magic": the same thirty-six rays that appear as the matter shell are the non-Clifford fuel for the processor.

## 6.4 The minimal classical shadow

The smallest classical headline compatible with the substrate is the pair

$$(\lambda, q) = (2, 3).$$

This is the same two-state, three-symbol signature made famous by the Wolfram–Smith weakly universal Turing machine. The present paper does not use that result as the proof of quantum universality; the proof is the Clifford closure plus magic/contextuality [14, 13]. The point is sharper: the classical shadow of the photonic holonet lands on the same minimality boundary. The quantum machine has forty rays and an $81 = q^4$ protected register, but its smallest visible finite-state control alphabet is already the substrate pair $2, 3$.

## 6.5 The Universality Theorem

**Theorem 6.1** (Clifford completeness from optics)**.** *The symplectic images of the photon's physical elements — the tritter $F_3$ on either register, the quadratic phase plate $S$, and the delay-conditioned* CX *— generate all of* $\mathrm{Sp}(4, \mathbb{F}_3)$ *(closure of order* 51840*, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases.* Witness: `bt825_universality_theorem.py`.

**Theorem 6.2** (Universality). *Clifford completeness, together with the machine's intrinsic magic supply (§9) and the strict positivity of its contextual fraction* ($1/10$, *exact*), *implies universal quantum computation by magic-state injection; for qutrits, contextuality is necessary and sufficient for magic distillation (Howard–Wallman–Veitch–Emerson) [3]. One self-entangled photon on the* $W(3,3)$ *mesh is a universal quantum computer whose transport is its gate action.*

## 7 Fractal scaling: the computer is the network

The single-core theorem is not the end of the architecture. The substrate is self-similar: any node may be replaced by a fresh copy of the same forty-node holonet, with the parent line/point incidence becoming the routing grammar for the child layer.

**Definition 7.1** (Recursive holonet). *Let* $\mathcal{H}_0$ *be one self-entangled photonic qutrit. Define* $\mathcal{H}_n$ *recursively by replacing each point of one copy of* $W(3,3)$ *with a copy of* $\mathcal{H}_{n-1}$*. Thus* $\mathcal{H}_1$ *is the single-core machine of this paper and* $\mathcal{H}_n$ *is a* $40$*-ary fractal computer/network.*

**Theorem 7.2** (BT827 holonet scaling law). *A level-n holonet has*

$$N_n = 40^n$$

*photonic leaf cores and*

$$I_n = 1 + 40 + \cdots + 40^{n-1} = \frac{40^n - 1}{39}$$

*total* $W(3,3)$ *instances. Its total edge-qutrit slots are* $240I_n$*, its directed transport slots are* $480I_n$*, its chart routers are* $540I_n$*, its apartment links are* $1620I_n$*, its mirror slots are* $2160I_n$*, and its local Clifford runtime atoms are* $51840I_n$*. Worst-case reversible routing across the recursive address uses at most*

$$8n = 8\log_{40} N_n$$

*moves, where* $8 = 3 + 5$ *is one in-chart hypercube diameter plus one chart-web apartment diameter. Durable commits use the distinct Császár/tomotope clock*

$$T(0) = 1, \qquad T(g) = 4(7^g - 1) \quad (g \geq 1).$$

Witness: `bt827_holonet_fractal_architecture.py`.

| level $n$ | leaves $40^n$ | $W(3,3)$ instances | mirror slots | runtime atoms | route bound |
|---|---|---|---|---|---|
| 1 | 40 | 1 | 2160 | 51840 | 8 |
| 2 | 1600 | 41 | 88560 | 2125440 | 16 |
| 3 | 64000 | 1641 | 3544560 | 85069440 | 24 |
| 4 | 2560000 | 65641 | 141784560 | 3402829440 | 32 |

This resolves the computer-engineering question. A conventional machine grows by adding processors connected by a separate network. The holonet grows by adding copies of the same object; each copy contains its own processor, router, memory, clock, and packet format. There is no von Neumann bus to saturate. At every scale, the edge set is both communication bandwidth and entangling-gate availability, the chart atlas is both routing table and instruction decoder, and the Steinberg 81-sector is both protected memory and fault-tolerant logical space.

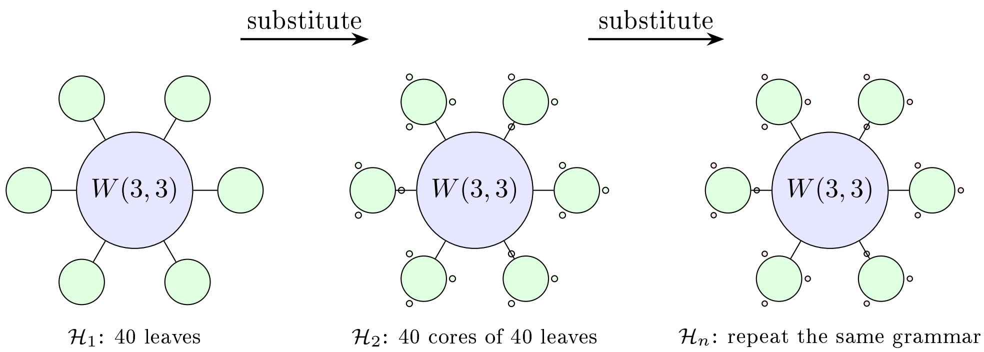


Figure 6: Fractal scaling schematic. The drawn fans show only a few children; the rule is exact substitution of forty child holonets at each substrate point. Routing grows logarithmically in the number of photonic leaves because each recursive digit costs at most eight reversible moves.

# 8 Runtime closure: compile, monitor, commit

The architecture now has an executable operating-system loop.

$$\text{address word} \longrightarrow \text{packet route} \longrightarrow \text{sentinel monitor} \longrightarrow \text{durable tomotope commit.}$$

BT828–BT831 make each arrow finite and testable.

**Theorem 8.1** (BT828 packet compiler)**.** *A recursive address word with digits in* $\{0, \ldots, 39\}$ *compiles to a sequence of digit packets*

$$(Q_3\text{-}XOR\ hops,\ apartment\ hops,\ D_{12}\ mirror\ slot,\ C_{12}\ phase,\ tomotope\ block).$$

*Each digit uses at most three XOR hops and at most five apartment hops, hence at most eight reversible moves. The stress route*

$$(0, 7, 14, 21, 28, 35) \longrightarrow (39, 32, 25, 18, 11, 4)$$

*compiles to* 47 *reversible moves, below the level-six bound* 48*.* Witness: `bt828_holonet_packet_compiler.py`.

**Theorem 8.2** (BT829 sentinel monitor)**.** *The projector onto the* $g = 15$ *adjacency sector of* $W(3, 3)$ *has exact entry profile*

$$P_{-4}(x, x) = \frac{3}{8}, \qquad P_{-4}(x, y) = \begin{cases} -\frac{1}{8}, & x \sim y, \\ \frac{1}{24}, & x \nsim y. \end{cases}$$

*Consequently a full context line* $K_4$ *is sentinel-invisible, while localized, mirror, and shell faults activate the sentinel with exact energies:*

| *fault* | *energy* | *normalized fraction* |
|---|---|---|
| *single point* | 3/8 | 5/13 |
| *adjacent edge* | 1/2 | 5/19 |
| *nonedge mirror pair* | 5/6 | 25/57 |
| *gauge shell* $N(x)$ | 6 | 5/7 |
| *matter shell* | 27/8 | 5/13 |

*The immune system is therefore not generic noise detection: it is blind to legal context-line traffic and tuned to off-context routing and shell congestion.* Witness: `bt829_fault_sentinel_monitor.py`.

**Theorem 8.3** (BT830 two-phase commit clock). *The runtime has two clocks. The reversible prepare phase finishes in at most $8n$ moves. The durable commit phase waits for*

$$T(n) = 4(7^n - 1).$$

*For $1 \le n \le 24$, $T(n)$ is always divisible by 24 and by 8, so it is aligned with both the local runtime stabilizer and the single-digit route window. But it is not always divisible by the full $8n$ route epoch: the first desynchronization is $n = 5$, with remainder $24 = f$. Thus the commit layer is deliberately slower and cover-like; it is not merely the route clock renamed.* Witness: `bt830_two_phase_commit_clock.py`.

## 8.1 The infinite-cover warning

The tomotope contributes a harder boundary than the earlier packet picture showed. Monson, Pellicer, and Williams prove that the tomotope has infinitely many distinct minimal regular covers; for coprime odd $p, q > 1$ there are minimal covers $R_p, R_q$ of the tomotope such that neither covers the other, and neither covers the special $R_2$ cover [17]. They also record the toroidal cover arithmetic

$$|\mathrm{Mon}(Q_k)| = 36864\, k^6, \qquad Q_k : \ (4, 24, 32, 8, 4)k^3$$

for vertices, edges, triangles, tetrahedra, and octahedra.

Architecturally this is a big correction. The 48-block tomotope middle layer is a stable local ABI, but the global regular cover is not unique. Durable storage must therefore carry a *cover index* $k$ as an implementation gauge. Fast routes compile to the same invariant packet, while persistent storage may choose one minimal cover family without invalidating the other non-comparable families. *Witness:* `bt831_tomotope_minimal_cover_architecture.py`.

## 8.2 Cover-indexed storage, rerouting, and guard bands

The infinite-cover warning would be useless as engineering unless it changed the runtime. BT832–BT834 make it operational. The principle is that the local tomotope packet is the ABI and the chosen regular cover is the storage backend.

**Theorem 8.4** (BT832 cover-indexed durable storage). *For every supported cover index $k$, durable storage is the lifted packet space*

$$\{0, \dots, 47\} \times \mathbb{Z}_k^3, \qquad |slots| = 48k^3,$$

*with projection back to the fast-route ABI by forgetting the $\mathbb{Z}_k^3$ coordinate. The kernel to the base tomotope packet has order*

$$k^6 = (k^3)^2,$$

*matching the Monson–Pellicer–Williams toroidal cover law $|\mathrm{Mon}(Q_k)| = 36864k^6$. Changing $k$ changes durable capacity and kernel size, but not the packet program seen by the compiler. Tested cover gauges $k = 3, 5, 7, 11, 13$ all reduce exactly to the same 48-block ABI, and larger $k$ reduces the load fraction of the same stress program.* Witness: `bt832_cover_indexed_durable_storage.py`.

**Theorem 8.5** (BT833 sentinel-aware packet rerouting). *The BT829 projector supplies a compiler-visible cost function. Before the durable commit, each digit route may insert up to two W33 waypoint points and minimize*

*sentinel energy first, reversible move count second.*

*The extra reversible moves are charged to the slower BT830 commit phase, not to the BT827 fast* $8n$ *route bound. In the level-six stress program the direct route uses* 47 *moves and total sentinel energy* 4*; the sentinel-aware route uses* 58 *moves, still far below the* 470592*-tick commit window, and lowers the total sentinel energy to* 2*. Each tested digit route strictly lowers its own* $g = 15$ *energy; adjacent endpoints can be context-completed to zero, and nonedge mirror pairs drop below their direct* $5/6$ *activation.* Witness: `bt833_sentinel_aware_packet_rerouter.py`.

**Theorem 8.6** (BT834 desynchronization guard-band arithmetic)**.** *The two-phase clock has an exact number-theoretic boundary. Full route-epoch synchronization is*

$$8n \mid T(n) = 4(7^n - 1),$$

*equivalently*

$$2n \mid 7^n - 1.$$

*For every odd prime power* $p^a \mid n$ *with* $p \neq 7$*, this requires* $\operatorname{ord}_{p^a}(7) \mid n$*; if* $7 \mid n$*, synchronization is impossible because the base is not coprime. The first failure is* $n = 5$*:* $\operatorname{ord}_5(7) = 4 \nmid 5$*, and the remainder is*

$$T(5) \bmod 40 = 24 = f.$$

*Thus the first cover-index where durable storage and the full route epoch separate is not empirical drift; it is the* $f = 24$ *guard band of the tomotope-backed runtime.* Witness: `bt834_desync_guard_band_arithmetic.py`.

**Theorem 8.7** (BT838 tomotope Wythoff runtime ladder)**.** *The Grünbaum–Coxeter/tomotope operation tables give the local tomotope counts*

$$\textit{original, rectified, truncated, maximal expanded, omnitruncated} \quad = \quad 4,\ 12,\ 24,\ 48,\ 96.$$

*These are exactly the runtime packet-growth layers:*

$$4 = \mu \quad (\textit{vertex carriers}), \qquad 12 = k \quad (\textit{edge axes}), \qquad 24 = f \quad (D_{12}\ \textit{lift}),$$

$$48 = 2f \quad (\textit{BT814 packet ABI}), \qquad 96 = 4f \quad (\textit{half-flag boundary}).$$

*The doubled omnitruncated boundary gives the verified full packet flags,* $2 \cdot 96 = 192$*. After a cover lift this becomes*

$$48k^3 = \textit{BT832 lifted packet capacity}, \qquad 192k^3 = |W_k|,$$

*the* $W_k$ *order already recorded in the tomotope cover architecture. Thus the Wythoff operations are not decorative variants; they are the compiler's packet expansion states:*

$$\textit{carrier} \to \textit{edge-axis} \to \textit{mirror lift} \to \textit{packet ABI} \to \textit{flag boundary}.$$

Witness: `bt838_tomotope_wythoff_runtime_ladder.py`.

## 9 The fuel: matter = magic

**Theorem 9.1** (The triality)**.** *In the photon's two-qubit reading, exactly the* 4 *basis rays are stabilizer states and all* 36 $\omega$*-rays are magic (no stabilizer state has support* 3*). Consequently three classifications coincide exactly, on rays* $[4, 36]$ *and on contexts* $\{4{:}1,\ 1{:}12,\ 0{:}27\}$*: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata.* The matter shell is the machine's non-classical resource sector. Witness: `bt822`.

**Theorem 9.2** (Three grades and the cube inside). *Stabilizer fidelity splits the* 36 *magic rays into grades*

$$\underbrace{8}_{2^q}\ \textit{deep}\ \Big(F=\tfrac{2+\sqrt3}{6}\Big),\qquad \underbrace{24}_{f}\ \textit{mid}\ \Big(F=\tfrac{5+2\sqrt3}{12}\Big),\qquad \underbrace{4}_{\mu}\ \textit{shallow}\ \Big(F=\tfrac34=\tfrac{q}{\mu}\Big),$$

*the shallow grade sitting exactly at the Werner decoherence threshold. The* 8 *deep rays are precisely the point set of a hyperbolic polar pair (the* 24*-cell vacuum axis), and the* 27 *fully-magic contexts stratify by grade signature as* $1+8+12+6$ *— the cell census of the cube.* Witnesses: `bt822`, `bt823`.

**Theorem 9.3** (Exact contextuality budget). *The maximum number of contexts a classical* $\{0,1\}$ *valuation can satisfy exactly-once is exactly* $36=(q!)^2$ *(complete branch-and-bound over all* $2^{40}$ *markings): classicality saturates at the spread count, the deficit is exactly* $\mu=4$*, and the contextual fraction is exactly* $1/10=1/\Phi_4$*. A perfect valuation would be an ovoid, and* $W(3,3)$ *has none: the true independence number is* $\alpha=7=\Phi_6$*, attained only by the beacon heptads of §12, against the Lovász bound* $\vartheta=10$ *— the* $\vartheta-\alpha$ *gap* $3=q$ *is the combinatorial face of Kochen–Specker.* Witnesses: `bt818`, `bt823`.

## 10 Memory, protection, and the immune system

**Theorem 10.1** (The code register is the Steinberg module). *The companion paper's CSS code* $[[240,81,4,3]]_3$ *stores its* 81 *logical qutrits in* $H_1$ *of the* $W(3,3)$ 2*-complex (*40 *vertices,* 240 *edges,* 160 *in-line triangles). Complete character computation over all* 25920 *group elements gives*

$$\langle \chi_{H_1},\chi_{H_1}\rangle = 1.$$

*Therefore* $H_1\otimes\mathbb{C}$ *is* irreducible *of dimension* 81*, hence equal to the unique* 81*-dimensional irreducible —* ***the Steinberg module*** *— with* $\chi_{H_1}(g)=\#\text{fixflags}-\#\text{fixpoints}-\#\text{fixlines}+1$ *(the Solomon–Tits alternating sum) verified for every* $g$*. The QECC logical space and the Schur-protected register below are one object: any equivariant logical error is a scalar, so only symmetry-breaking noise can touch the code, and the sentinel/clock layers are exactly the symmetry monitors. The homological dictionary then completes:* $H_0$ *is trivial (the vacuum pole),* $H_1$ *is Steinberg (the register), and* $H_2$ *(dim* 40*, one tetrahedron-boundary* 2*-cycle per line) is the* sign-twisted *line module — a fixing symmetry acts on a line's* 2*-cycle by the parity of its induced* $S_4$ *action (verified for all* 25920 *elements;* not *the plain permutation module): the top homology is the* oriented *timetable carrier, feeding the chirality ledger.* Witnesses: `bt861_code_register_is_steinberg.py`, `bt862_h2_is_the_line_module.py`.

**Corollary 10.2** (Three generations from Steinberg vanishing). $\chi_{\mathrm{St}}(g)=0$ *iff* $3\mid \mathrm{ord}(g)$ *(verified for all* 25920 *elements). Hence* any *order-*3 *symmetry of the substrate splits the matter register into eigenspaces of exactly* $27+27+27$ *— the three-generation decomposition is forced for every choice of triality element, not selected — and every order-*9 *element (*5760 *of them) refines it into nine* 9*-dimensional sub-generations. In the code's own characteristic the parameters survive:* $\mathrm{rank}_3\partial_0=39$*,* $\mathrm{rank}_3\partial_1=120$*,* $\dim H_1(\mathbb{F}_3)=81$*, so* $[[240,81,4,3]]_3$ *is exact over* $\mathbb{F}_3$*. The number of fermion generations is the defining-characteristic degeneracy of the protected register.* Witness: `bt863_generations_from_steinberg_vanishing.py`.

**Theorem 10.3** (The dual-torsor Steinberg compiler). *Let* $H_{27}=3_+^{1+2}$ *be the canonical Heisenberg* $O_3$ *of the point parabolic and let* $A_{27}=\mathbb{F}_3^3$ *be the canonical translation* $O_3$ *of the Bell-line parabolic. Their* 27*-point and* 27*-line shells are regular torsors (BT858). On the protected register,*

$$\mathrm{St}{\downarrow_{H_{27}}}=3\,\mathbb{C}[H_{27}],\qquad \mathrm{St}{\downarrow_{A_{27}}}=3\,\mathbb{C}[A_{27}],$$

*because both restricted characters are exactly* $\{81^1, 0^{26}\}$*. The statement survives in the code's own field and is constructive:*

$$H_1(W(3,3);\mathbb{F}_3){\downarrow_{H_{27}}}\cong \mathbb{F}_3[H_{27}]^{\oplus 3}, \qquad H_1(W(3,3);\mathbb{F}_3){\downarrow_{A_{27}}}\cong \mathbb{F}_3[A_{27}]^{\oplus 3}.$$

*For each torsor the verifier finds three cycle seeds whose* 27 *group translates add* $27+27+27$ *independent classes modulo the* 120*-dimensional boundary space. Thus the same logical memory has a flat program basis and a curved state basis:* $A_{27}$ *is the commuting three-trit address compiler, while* $H_{27}$ *is the noncommuting qutrit-phase compiler.*

*The center* $Z(H_{27}) = \langle z\rangle \cong C_3$ *selects a canonical triality. Over* $\mathbb{C}$ *its central-character blocks are*

$$H_1 = H_{1,1}\oplus H_{1,\omega}\oplus H_{1,\omega^2}, \qquad \dim H_{1,j} = 27.$$

*The trivial block contains the nine linear characters with multiplicity* 3*; the two nontrivial blocks contain the conjugate* 3*-dimensional Schrödinger representations with multiplicity* 9*. Over* $\mathbb{F}_3$*, where* $x^3-1=(x-1)^3$*, the phases coalesce into a canonical unipotent flag. With* $N = z - I$*,*

$$(\operatorname{rank} N, \operatorname{rank} N^2, \operatorname{rank} N^3) = (54, 27, 0),$$

*so*

$$0 \subset \ker N \subset \ker N^2 \subset H_1, \qquad \dim = (0, 27, 54, 81),$$

*has three successive quotients of dimension* 27*. Equivalently,* $z$ *acts by* 27 *Jordan blocks of length* 3*. Complex generations are phase sectors; native qutrit generations are the three layers of one depth-*3 *memory stack.*

*The freeness is canonical, but the displayed orbit bases are not:* $H_{27}$ *and* $A_{27}$ *are nonisomorphic, and a basis-to-basis matrix still requires a chosen point, line, shell origin, and cycle seeds. No canonical intertwiner is claimed.* Witness: `bt865_dual_torsor_steinberg_compiler.py`.

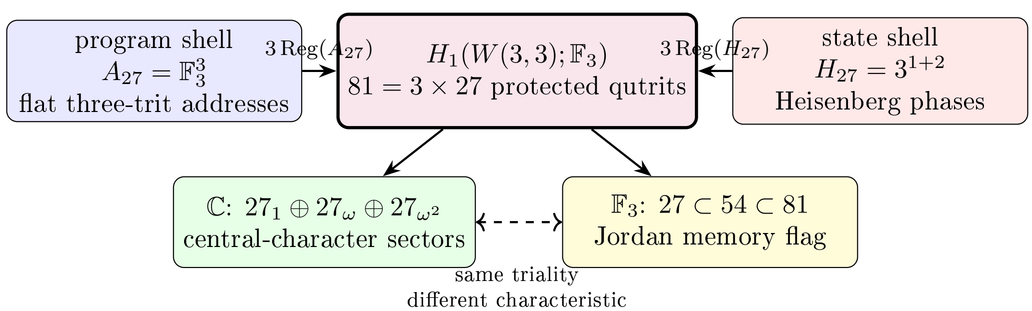


Figure 7: The dual-torsor compiler. Flat program translations and noncommutative state translations give two free rank-3 module coordinates on one Steinberg-protected register. The Heisenberg center reads as three phase sectors over $\mathbb{C}$ and as a three-step unipotent memory flag over $\mathbb{F}_3$.

**Theorem 10.4** (Steinberg memory)**.** *The Levi-graph cycle space and the chart-overlap* 81*-sector are both irreducible and equal to the Steinberg representation of* $\mathrm{PSp}(4,3)$ *— the Solomon–Tits homology of the building, whose canonical basis is the* 81 *apartments through any chamber. Schur's lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow* 3*-subgroup the module is free of rank one: the protected memory is one free generator over the substrate's ternary symmetry.* Witnesses: `bt742`, `bt744`.

**Corollary 10.5** (The oriented timetable spectrum). *Let $P = 3^3{:}S_4$ be the index-40 line parabolic. The ordinary line permutation module and the oriented top-homology module are the two inductions of its two linear characters:*

$$\mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(1) = 1 \oplus 15 \oplus 24, \qquad H_2 = \mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(\mathrm{sign}_{S_4}) = 5_\omega \oplus 5_{\omega^2} \oplus 30.$$

*The two 5-dimensional characters are complex conjugates over $\mathbb{Q}(\zeta_3)$; the 30 is rational. Under the outer extension $W(E_6) = U_4(2){:}2$, the conjugate pair is the restriction of one irreducible 10-dimensional representation, whereas the 30 has two inequivalent extensions. Thus the oriented timetable header reads as a Weyl-visible chiral channel* 10 *plus a payload* $30^{\pm}$ *whose outer parity is not fixed by the projective module alone.*

*The completed homology spectrum is*

$$H_0 = 1, \qquad H_1 = \mathrm{St}_{81}, \qquad H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30,$$

*and resolves the Euler charge representation by representation:*

$$1 - 81 + (5 + 5 + 30) = -40 = -v.$$

*BT867 resolves the tempting cache comparison negatively: neither* 162*-cache branch is one of the two 5-dimensional sectors. The branches are isomorphic copies with identical character overlaps; the outer involution acts on their multiplicity label. The possible identification of the two 30-extensions with BT857's local gauge bit remains open.* Witness: `analysis/bt866_h2_oriented_irreducible_decomposition.py`.

**Theorem 10.6** (The cache/transport extension boundary). *Let $H = 3^3{:}S_4$ be the order-*648 *Bell-line parabolic. The two* 162*-slot BT859 cache branches have conjugate cyclic stabilizers and are therefore isomorphic transitive $H$-sets,*

$$\mathcal{C}_L \cong H/C_4 \cong \mathcal{C}_R.$$

*Their permutation characters are equal. In particular, their scalar products with the restrictions of the three BT866 characters $(5_\omega, 5_{\omega^2}, 30)$ are both $(1, 1, 8)$: each cache sees both conjugate 5-sectors equally. "Left" and "right" are copy labels, not internal spectral chiralities.*

*The parabolic contains one conjugacy class of* 81 *cyclic $C_4$ subgroups, and*

$$N_H(C_4) = D_8, \qquad H/C_4 \longrightarrow H/D_8, \qquad 162 \longrightarrow 81$$

*is the canonical free two-sheet cache cover. Both copies cover this same stabilizer base. On their union GAP computes*

$$C_{\mathrm{Sym}(324)}(H) \cong D_8, \qquad 324 = 81 \times 4 = 81 \times 2_{\mathrm{deck}} \times 2_{\mathrm{cache}},$$

*with exactly* 81 *commutant orbits of size four. Thus the four-state fiber carries the symmetry of a square, not merely a numerical pair of bits.*

*This cache carrier is not the older ternary* 162*-sector. On one cache fiber the deck swap and its two projectors are*

$$D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\pm} = \frac{I \pm D}{2}, \qquad \mathbb{F}_3^2 = \operatorname{im} P_+ \oplus \operatorname{im} P_-.$$

*Since 2 is invertible in $\mathbb{F}_3$, the cache splits; indeed $(D-I)^2 = D-I$. By contrast, packet transport uses*

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0, \qquad (I+N)^3 = I,$$

*and tensors this indecomposable fiber with the 81-register to obtain the non-split sequence $0 \to 81 \to 162 \to 81 \to 0$. No change of basis can identify the idempotent cache difference with the nilpotent transport shift. The architecture therefore separates a reversible* address/control plane *from a memory-bearing ternary* temporal data plane *by extension class, not by convention.* Witness: `analysis/bt867_cache_split_transport_nonsplit_boundary.py`.

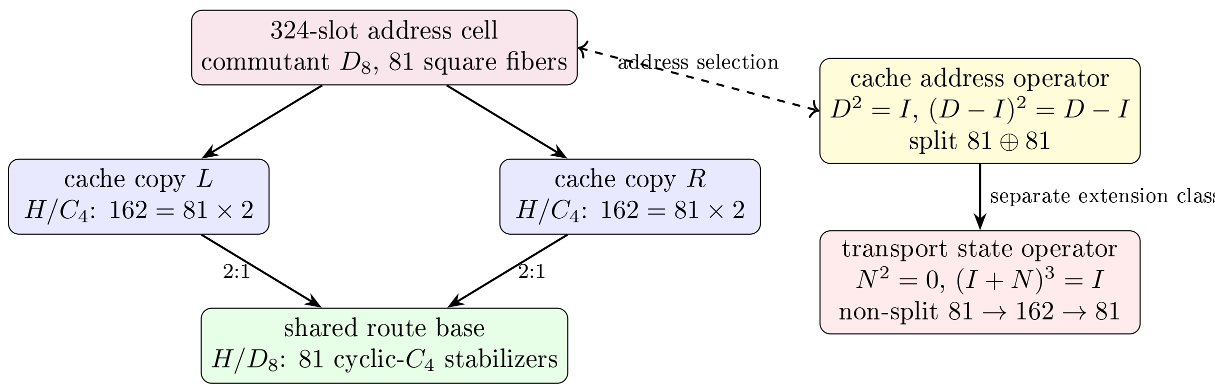


Figure 8: The local cache cell and the split/non-split boundary. The $D_8$ square fiber chooses a reversible address over one of 81 routes; the independent ternary nilpotent updates state while retaining temporal memory. Equal 162-dimensional counts do not imply equivalent modules.

The error layer is native: the $[[240, 81, 4, 3]]_3$ CSS code on the edge qutrits has all four parameters substrate-primitive; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web's unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

## 11 Time: three clocks and a quasicrystal

**Theorem 11.1** (Clock trichotomy)**.** *Of the program's three natural clocks, exactly one is internal: $\mathbb{Z}_{12}$ (the rectangle clock, $12 \mid 51840$) runs the selector and duo machinery, while $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) do not divide the group order and act as external references — both with decimal period $6 = q!$, so the digit expansion of $1/7$ is literally the internal clock's shadow modulo its duo bit.* Witnesses: `bt774`, `bt803`, `bt807`, `bt819`.

**Theorem 11.2** (The Boerdijk–Coxeter time quasicrystal)**.** *Driving the loop by the BC twist $\theta =$* $\arccos(-2/3)$ *per round trip yields a stroboscopic orbit that provably never repeats (Niven), with the Steinhaus three-distance signature at all substrate step counts and — the substrate's signature moment — exactly* two *gap lengths at $n = 30 = h(E_8)$, the BC ring length, where closure happens in $S^3$ (the 600-cell) rather than the phase circle (deficit* 0.0158*). This is the photonic sibling of the Fibonacci-drive dynamical topological phase.* Witness: `bt820`.

**Theorem 11.3** (The arrow of time)**.** *Without its two-trit feedforward record, the photon's self-configuration loop is the completely depolarizing channel with unique fixed point $I/q$ — exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at* $\log_3 q^2 = 2$ *trits per cycle.* Witness: `bt823`.

## 12 Synchronization: beacons and schedules

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete $K_7$ interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3,3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses: `bt817`, `bt819`.*

**Theorem 12.1** (Schedule-overlap law)**.** *Two distinct schedules share exactly* 1 *or exactly* $4 = \mu$ *contexts: each schedule has* 15 near *partners (overlap* 4*) and* 20 far *partners (overlap* 1*), the two relation classes of the double-six diagonal* $[1, 15, 20]$*, with the exact counting identity* $1 \cdot 20 + 4 \cdot 15 = 80 = 10 \times (9 - 1)$*. Operationally: a cheap timetable switch preserves* 4 *of* 10 *calibrated contexts and a far switch only* 1*, so a controller hopping among near partners re-calibrates* 6 *contexts per hop — the timetable library is itself a metric space whose geometry is the double-six relation.* Witness: `bt835_schedule_overlap_theorem.py`.

**The Grünbaum–Coxeter cells.** Each schedule also has an internal shape. The full schedule stabilizer $S_6$ is 2-homogeneous on the 10 contexts, but its icosahedral core $A_5$ — the 600-cell level of the platonic ladder, and *exactly* the automorphism group of the hemi-dodecahedron $\{5,3\}_5$ — splits the $\binom{10}{2} = 45$ context pairs as $45 = 15 + 30$, and the 15-orbit graph on the 10 contexts is the **Petersen graph**: the edge skeleton of the hemi-dodecahedron, the cell of Coxeter's exceptional 57-cell. Dually, the hidden 6-set of the $S_6$ action realizes the hemi-icosahedron $\{3,5\}_5$ — the cell of Grünbaum's 11-cell — with incidence count $(6, 15, 10)$: six hidden points, fifteen duads, ten triple-splits = the contexts themselves. The exceptional polytopes' groups do not embed in $\mathrm{Sp}(4, \mathbb{F}_3)$, but they are substrate arithmetic ($|\mathrm{PSL}(2, 11)| = 660 = k \cdot N_{\mathrm{eff}}$, $|\mathrm{PSL}(2, 19)| = 3420 = k \cdot g \cdot 19$, Heawood genus $H(19) = 20$ = the BC ring count, and $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$): the full 11-cell and 57-cell live elsewhere, yet their *cells* are carried by every measurement timetable of this machine, glued by the substrate's own group instead of $\mathrm{PSL}(2, p)$. *Witness: `bt836_gc_hemicells_in_spreads.py`.*

**Theorem 12.2** (The library is a classical geometry)**.** *The timetable library carries its own rank-3 geometry, and the hemicell structure fibers over it with perfect uniformity: (i) every skew line pair lies in exactly* 3 *schedules, so the* (schedule $\supset$ pair) *flag count is* $36 \times 45 = 1620$ *— the apartment count of the Tits building — and* $540 \times 3$*; (ii) the near-partner graph on the* 36 *schedules is strongly regular* $\mathrm{SRG}(36, 15, 6, 6)$*, the* other *classical rank-3 graph of* $U_4(2) \cong \mathrm{PSp}(4, \mathbb{F}_3)$*, so the control plane (timetables) and the data plane (*$W(3,3) = \mathrm{SRG}(40, 12, 2, 4)$*) are the two rank-3 geometries of one group; (iii) each schedule stabilizer contains exactly* 6 *icosahedral* $A_5$ *cores (*$216 = 6^3$ *in all, each marking its unique schedule via its* $[10, 30]$ *line orbits) giving* 6 *distinct Petersen splits; each internal pair is a Petersen edge under exactly* 2 *of the* 6 *cores, hence every skew pair of* $W(3,3)$ *has exactly* $6 = 3 \times 2$ *hemi-dodecahedral homes (*$3240 = 540 \times 6$ *Petersen flags).* Witness: `bt837_schedule_library_geometry.py`.

**Theorem 12.3** (Two compasses, twelve Petersens, $4 \cdot K_{10}$)**.** $\mathrm{PSp}(4, \mathbb{F}_3)$ *has exactly two conjugacy classes of* $A_5$*, both with normalizer* $S_5$ *and* 216 *conjugates, unfused by the outer automorphism, and* both fiber over the same 36 schedules*: every schedule stabilizer contains* $12 = 6 + 6$ *icosahedral cores —* 6 duad *cores (line signature* $[10, 30]$*, the Theorem-12.2(iii) cores) and* 6 pentad *cores (signature* $[5, 5, 10, 20]$*: transitive on the schedule, plus two distinguished* pentads *of* 5 *pairwise-disjoint lines outside it). The two* 216*-sets are non-isomorphic* $G$*-sets (rank* 10 *suborbits* $[1,5,10,10,20,20,20,30,40,60]$ *vs* $[1,5,10,10,20,20,30,30,30,60]$*); the Petersen flags form the coset geometry* $\mathrm{PSp}(4,\mathbb{F}_3)/D_4$ *(*3240 *flags, dihedral stabilizer). There are* 432 *distinct pentads in* two chiral orbits *of* 216 *(stabilizer order* 120*); every core pairs one LEFT with one RIGHT pentad, and each pentad serves exactly one core. Pentads are* maximal partial spreads*: no schedule contains one. The two pentads of one core interlock as* $K_{5,5}$ *minus a perfect matching — mutual disjointness is* forbidden *by the overlap law (it would create a* 0*-overlap schedule) — and the deleted matching consists of* 5 skew *pairs, i.e. five hypercube charts; over the* 216 *cores these matchings cover the* 540*-chart atlas* ***exactly twice*** *(*$1080 = 540 \times 2$*): the routing fabric is readable off the pentad compasses, five charts per needle. Finally,* all twelve *cores split the* 45 *internal pairs as Petersen* 15*-orbits, and the twelve Petersen edge-sets cover every pair exactly* $\mu = 4$ *times:*

$$4 \cdot K_{10} \;=\; 12 \textit{ Petersens.}$$

*Schwenk's classical obstruction (no edge-partition of* $K_{10}$ *into* 3 *Petersen graphs) dissolves at the substrate multiplicity* $\mu$*. The numerical coincidence* $3240 = \#\{\textit{triangles of } Q\}$ *is refuted at the* $G$*-set level (orbits* $[360, 2880]$ *vs transitive).* Witnesses: `bt843_icosahedral_compass.py`, `bt844_double_five_and_six_petersens.py`, `bt845_pentad_exchange_and_chart_double_cover.py`.

**Theorem 12.4** (The amalgamation ladder)**.** *The* 11*-cell and* 57*-cell are* universal *abstract polytopes: the unique ones with hemi-icosahedral facets and hemi-dodecahedral vertex figures (group* $L_2(11)$*, no proper quotients) and vice versa (*$L_2(19)$*). Their incidence is icosahedral-subgroup geometry: each of* $L_2(11)$*,* $L_2(19)$*,* $U_4(2)$ *has exactly two conjugacy classes of* $A_5$*, and in all three the distinguished incidence is* intersection in $D_{10}$ *with valency* 6 *— the* 11*-cell's vertex–facet relation, the* 57*-cell's facet adjacency (the Perkel graph), and the substrate's compass incidence (*$216+216$*, bipartite* 6*-regular,* $216 \times 6 = 1296 = 6^4$ *incident pairs* $=$ 36 *schedules* $\times$ 36 *core pairs). The closures of one amalgam* $A_5 *_{D_{10}} A_5$ *(GAP):*

$$\langle A, B\rangle = \begin{cases} L_2(9) \cong A_6 \ (360) & \textit{in } U_4(2)\textit{: inside exactly one schedule stabilizer,}\\ L_2(11) \ (660) & \textit{the } 11\textit{-cell,}\\ L_2(19) \ (3420) & \textit{the } 57\textit{-cell.}\end{cases}$$

*One local icosahedral amalgam, a* $\mathrm{PSL}(2)$*-ladder of completions over* $9 = q^2$*,* $11 = k-1$ *(the Ihara prime),* 19 *(Heawood,* $H(19) = 20$*). Universality plus Lagrange (*$11, 19 \nmid 25920$*) shows the substrate carries the* complete local data *of both exceptional polytopes while their rank-*4 *amalgamation is arithmetically forbidden inside* $\mathrm{Sp}(4,\mathbb{F}_3)$*: the GC polytopes are the substrate's sibling completions, not subobjects. One step up the universal ladder (*$\{\{5,3\},\{3,5\}_5\}$*,* 10006920 *facets) the completion group is* $J_1 \times L_2(19)$ *— the first sporadic Janko group, whose involution centralizer* $2 \times A_5$ is *the compass datum. Finally, the compass Petersen* 15*-orbit carries exactly* 12 *pentagons splitting under the compass into two* 6*-orbits, each covering every edge exactly twice: two* chiral fully-faced *hemi-dodecahedra per needle, not merely skeletons. The dark sector closes the calculus: its* 190 *line pairs split* $[10,30,30,30,30,60]$ *under the core into a perfect matching of* 10 dark charts*, two chiral* $5 \times K_4$ *partitions into skew tetrads, the two dark dodecahedra, and a* 6*-regular meeting frame; the dark matching is* schedule-shadow pairing *(each dark line meets exactly* 4 *schedule lines; matched lines share their shadow), and in* $K_5$ *terms the ten shadows are exactly the ten "edge + opposite triangle" configurations — a canonical bijection dark chart* $\leftrightarrow$ $K_5$ *edge* $\leftrightarrow$ *schedule line. The timetable is stored three ways (lit-chart transversals,* $K_5$ *edges, dark shadows): built-in erasure redundancy. Each chiral tetrad's four distinguished edges form a* $K_5$ star*, the five tetrads of each partition marking the five lit charts exactly once:* $K_5$ *vertices are stored three ways (lit charts, left/right tetrads), edges* *two ways (schedule lines, dark charts) — every $A_5$-orbit structure of the pentad core except the dodecahedra and the meeting frame is a $K_5$ element — and those two fold in as well: each chiral dark dodecahedron is the* antipodal double cover *of the Petersen (Kneser) graph on the* 10 *dark charts, with deck transformation the shadow matching (antipode = shadow partner, verified), and the meeting frame is the doubled Johnson graph $T(5)$. The $K_5$-relation census is constant on every orbit (matching = same edge, tetrads and meeting = adjacent, dodecahedra = disjoint): the dark sector is the canonical* 2:1 *cover of the $K_5$ edge-world, with the Petersen graph appearing three ways in one needle (schedule skeleton, dark Kneser quotient, doubled chiral covers).*

**Theorem 12.5** (The dark charts are the mirror bus)**.** *The* (core, dark chart) *slot space —* 216 *pentad cores* $\times$ 10 *dark charts* = 2160 *slots, with every skew pair occurring as a dark chart in exactly* 4 *cores (*2160 = 540 $\times$ 4*, the repair-atlas factorization) — is **G-set isomorphic to the** 2160**-slot** $D_{12}$ **mirror bus** of the middleware: the slot stabilizer has order* 12 *with the $D_{12}$ profile* $\{1{:}1, 2{:}7, 3{:}2, 6{:}2\}$ *and is* conjugate in $\mathrm{PSp}(4, \mathbb{F}_3)$ *to the antipode-slot stabilizer (direct search over all* 25920 *elements). The multiplicity-*4 *agreement is canonical: a dark chart's transversal tetrad is its schedule shadow, so the* 4 *host cores' schedules all contain the pair's* 4 *transversals. Hence the compass layer* generates *the middleware — timetables from its lit side (reconstruction), the transport bus from its dark side — and the mirror bus acquires a hardware identification: one slot = one needle's dark chart.* Witnesses: `bt856_dark_mirror_bus.py`; further: `bt853_dark_orbit_zoo.py`, `bt854_dark_chart_transversal_duality.py`, `bt855_dark_sector_k5_complete.py`.

Witnesses: `bt848_universal_amalgam_ladder.py`, `.tmp/gap_bt848*.g`; cf. Hartley's classification (17 universal rank-4 locally projective polytopes, 441 quotients) and Hartley–Leemans on $\{5, 3, 5\}$.

**The barren rung and the maniplex workaround.** Relaxing polytopes to *maniplexes* (flag graphs without the intersection property; the tomotope middleware of this machine is a uniform 4-polytope whose monodromy $18432 = 96 \times 192$ fails that property, giving it *infinitely many* distinct minimal regular covers — the canonical rank-4 cover pathology) does not lift the obstruction: exhaustive search shows $U_4(2)$ admits *no* rank-4 regular maniplex with a 5 in its type, and the amalgam closure $A_6 = L_2(9)$ admits no regular maniplex at any rank — it is the *barren rung* of the PSL(2) ladder, whose populated rungs are exactly the 11-cell ($q$=11), the 57-cell ($q$=19), and Leemans–Toledo's degenerate single-facet $\{5, 5, 5\}$ family (our control search rediscovers all of them, including both exceptional polytopes). This is why the Grünbaum–Coxeter content of the substrate is *delocalized* over the 36 schedules — Petersen splits, chiral pentads, chart covers, dark dodecahedra — rather than crystallized in one flag object, and why the machine's runtime uses the tomotope: it is the flag structure that survives the obstruction. The tomotope's symmetry type is now exact: $|\mathrm{Aut}| = 96$ (independent centralizer computation on the 192 flags), two flag orbits $[96, 96]$, **class** $2_{\{0,1,2\}}$ — only the cell-color crosses orbits, and the orbits are precisely "flag in a tetrahedron" vs "flag in a hemioctahedron": the middleware *alternates spherical and projective cells*, the exact situation the classical "locally $X$" taxonomy cannot name, and the setting of Mochán's 2-orbit polytopality theory. Moreover the tomotope is a **Cayley maniplex over** $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ (vertex action = full $S_4$ with a regular Klein subgroup; edge and face actions faithful and transitive): the middleware's vertex layer is a free $\mathbb{F}_2^2$-torsor, so register XOR-addressing extends from the chart atlas into the runtime with no added structure. Reading Hartley's full classification (seventeen universal locally projective rank-4 polytopes) against the substrate adds three facts: Aut(**tomotope**) **is itself a universal polytope group** — isomorphic to $\Gamma(\{\{4, 3\}_3, \{3, 4\}_3\})$, the quotient-free doubly-projective $\{4, 3, 4\}$ universal built from hemicubes and hemicrosses (the tomotope's own projective cell type), with the case-10 group = $C_2 \times$ Aut of order 192 = the flag count; the mixed

$\{3,5,3\}$ amalgam (icosahedral facets, hemi-dodecahedral vertex figures) *does not exist* — the construction closes into the 11-cell — while the dodecahedral mix is the $J_1$ giant; and the chart–compass universal $\{\{4,3\},\{3,5\}_5\} = 2^{\mathcal{I}}$ (cube facets = $Q_3$ charts, hemi-icosahedral vertex figures = compass cells, vertices = $\mathbb{F}_2^6$ on the hidden 6-set) has order $3840 \nmid 25920$: blocked like the rest of the ladder. *Witnesses:* `bt849_maniplex_barren_rung.py`, `bt850_tomotope_two_orbit_class.py`, `bt851_tomotope_cayley_test.py`, `bt852_seventeen_universals.py`, `.tmp/gap_bt849*.g`, `.tmp/gap_bt852*.g`.

**Reconstruction, the chart $K_5$, and the dark dodecahedra.** The pentad core's full signature $[5,5,10,20]$ (lines), $[20,20]$ (points) is rigid: both chiral pentads cover the *same* 20-point orbit (lit/dark halves of the substrate), the five matching charts' common-transversal bundles overlap to exactly the 10 schedule lines — **a pentad pair generates its timetable** — and the line $\to$ chart-pair map is a bijection onto the $\binom{5}{2} = 10$ edges of $K_5$: the schedule *is* a $K_5$ on its own charts (4 transversals per chart, 2 charts per line). The 20 dark lines complete the $\{5,3\}$ family: their 190 pairs split $[10,30,30,30,30,60]$ under the core, and exactly two of the 30-orbits are **dodecahedron skeletons** (3-regular, girth 5, diameter 5) — a chiral pair of genuine dodecahedra beneath every compass needle, one level under the hemi-dodecahedral Petersen splits of the schedule itself. *Witnesses:* `bt846_pentad_core_anatomy.py`, `bt847_dark_dodecahedron.py`.

**Theorem 12.6** (BT839 GC operation Euler/flag audit)**.** *Treating the full Grünbaum–Coxeter note as a data table gives a second invariant. The base tables for the* 11*-cell,* 57*-cell, tomotope partial a, and tomotope partial b are all Euler-neutral:*

$$V - E + F - C = 0.$$

*But every Wythoff operation table has a fixed family charge:*

$$\chi_{\mathrm{op}}(11\text{-}cell) = -11, \qquad \chi_{\mathrm{op}}(57\text{-}cell) = -57, \qquad \chi_{\mathrm{op}}(tomotope) = -4.$$

*The omnitruncated rows are full-flag carriers:*

$$660 = |\mathrm{PSL}(2,11)| = 11 \cdot A_5 = kN_{\mathrm{eff}},$$

$$3420 = |\mathrm{PSL}(2,19)| = 57 \cdot A_5 = k \cdot g \cdot 19,$$

*and the tomotope has* 96 *half-flags, whose double is the verified* 192*-flag BT814 packet. The* 57*-cell also locks to the W33 timetable library: BT837 gives* 3240 *Petersen homes, and*

$$3240 + k \cdot g = 3240 + 180 = 3420.$$

*Thus one $k \cdot g$ sentinel sheet completes the W33 Petersen-home carrier to the* 57*-cell flag count.*

*The regular-polychoron alignment supported by the cell/symmetry data is*

$$11\text{-}cell \to 600\text{-}cell, \qquad tomotope \to 24\text{-}cell, \qquad 57\text{-}cell \to 120\text{-}cell.$$

*The* $57 \to 120$ *link is direct at the dodecahedral cell level; the* $11 \to 600$ *link is weaker but real, through hemi-icosahedral local structure and the quotient* $2I \to A_5$ *rather than through a direct tetrahedral-cell match. The swapped* $11/57$ *orientation is recorded as a lower-scoring dual/vertex-figure hook, not discarded.* Witness: `bt839_gc_operation_euler_flag_audit.py`.

**Theorem 12.7** (BT840–BT842 carrier closure)**.** *The three unresolved carriers in the Grünbaum–Coxeter paragraph are now explicit objects.*

***(i) The* 180 *sentinel sheet.*** *The verified Clifford $L/R$ boundary has* 36 *cells arranged as a* $6 \times 6$ *rook grid. Its zero-overlap relation is not a numerical afterthought: it is the union of the* 6 *row fibres and* 6 *column fibres, each fibre carrying the duads of a six-set. Hence*

$$12 \cdot \binom{6}{2} = 12 \cdot 15 = k \cdot g = 180.$$

*This is exactly the sheet missing from BT837:*

$$3240 + 180 = 3420 = |\mathrm{PSL}(2, 19)|.$$

*Equivalently, the timetable library gives* 216 *Petersen cores (*$216 \cdot 15 = 3240$*), and the rook boundary supplies the missing* 12 *six-set fibres (*$12 \cdot 15 = 180$*).*

***(ii) The* 660 *carrier.*** *At an apex cell* $(r, c)$ *of the same* $6 \times 6$ *grid, remove the incident row and column. The apex plus the five nonincident rows and five nonincident columns gives*

$$11 = 1 + 5 + 5.$$

*Crossing these eleven labels with the verified* 60*-element Clifford* $A_5$ *selector gives*

$$11 \cdot A_5 = 11 \cdot 60 = 660 = kN_{\text{eff}}.$$

*This is a carrier theorem only: it does not assert an internal* $\mathrm{PSL}(2, 11)$ *action on* $W(3, 3)$*.*

***(iii) The* 96 *tomotope half-flags.*** *In the* $D_4$*-root model of the* 24*-cell, every edge lies in a unique central hexagon plane. The two endpoint axes determine the third, missing axis in that plane, so each edge maps to a Reye incidence* $(\textit{missing axis}, \textit{hexagon plane})$*. Each of the* 48 *Reye incidences has exactly two edge lifts:*

$$96 = 2 \cdot 48.$$

*Thus the tomotope omnitruncated half-flag count is literally the* 24*-cell edge lift of the common tomotope/Reye spine.*

*The hexagon clue supplies the conceptual bridge to the* 11*-cell. Each central* 24*-cell hexagon is a six-root cycle* $C_6$*. Completing that cycle to a full* $K_6$ *gives* 15 *duads, exactly the hemi-icosahedral skeleton count of the* 11*-cell cell. Across the* 16 *central hexagons,*

$$16 \binom{6}{2} = 16 \cdot 15 = 240,$$

*the W33 edge count and the* $E_8$ *root count. The tested dot profile of these* 240 *duad slots is*

$$96_{\langle x,y\rangle=1} \;+\; 96_{\langle x,y\rangle=-1} \;+\; 48_{\langle x,y\rangle=-2},$$

*namely live* 24*-cell edges, mirror pairs, and repeated opposite-axis Reye incidences.* Witnesses: BT840, BT841, and BT842; executable scripts are in `analysis/`.

# 13 Build sheet and falsifiable witnesses

**Components** (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

**Witnesses** (kill criteria, all parameter-free):

| witness | predicted value | substrate form |
|---|---|---|
| trace–Choi visibility of $F_3$ | 1/3 | $1/q$ |
| trace–Choi visibility of $X, Z$ | 0 | — |
| Kochen–Specker classical budget | 36/40 | $(q!)^2/v$ |
| beacon-mesh pair visibility | 1/3 (all 21 pairs) | $1/q$ |
| Werner separability threshold | 3/4 | $q/\mu$ |
| BC-drive gap census at $n$=30 | 2 gap lengths | $h(E_8)$ ring |
| Hashimoto transport phases (one tick) | $\arctan\sqrt{10},\ \pi - \arctan(\sqrt{7}/2)$ | $\Phi_4, \Phi_6$; $\lvert u\rvert^2 = 11$ |
| key-agreement rate | 13/40 | $\Phi_3/v$ |
| $D_{12}$ mirror-slot stabilizer | order 12 | projective bus |
| runtime factorization | $24 \cdot 45 \cdot 48$ | $\lvert\mathrm{Sp}(4, \mathbb{F}_3)\rvert$ |
| one recursive route digit | $\leq 8$ reversible hops | $3 + 5$ |
| level-$n$ leaves | $40^n$ | $v^n$ |
| level-$n$ $W(3,3)$ instances | $(40^n - 1)/39$ | geometric tower |
| durable commit clock | $4(7^g - 1)$ ticks | tomotope/Csaszar |
| compiled stress route | $47 < 48$ moves | BT828 level six |
| cover-lifted packet capacity | $48k^3$ slots | BT832 storage gauge |
| Bell–compass prefix router | $648 + 324 + 162 + 162$ | code $0, 10, 110, 111$ |
| sentinel-aware stress energy | $4 \to 2$ | BT833 reroute |
| full route sync criterion | $2n \mid 7^n - 1$ | BT834 guard band |
| tomotope Wythoff runtime ladder | $4, 12, 24, 48, 96$ | BT838 packet growth |
| GC operation Euler charges | $-11, -57, -4$ | BT839 table audit |
| sentinel projector trace | 15 | $g$ sector |
| first full-route desync | $n = 5$ remainder 24 | $f$ guard band |
| tomotope cover ABI | 48 blocks / 192 flags | cover-invariant |

A measured deviation in any row falsifies the corresponding layer.

## 14 Implications, corrections, and the ethos

This section states what the architecture means — for physics, for computing, for networking, and beyond the laboratory. Every claim below is either (i) a theorem with an executable witness in this paper's ledger, or (ii) an implication of such a theorem, flagged as such. Nothing here floats free of the verification chain.

### 14.1 Physics

**The vacuum is a gauge choice among five.** The decomposition $40 = 1 + 12 + 27$ (self, gauge, matter) that organizes the entire Standard-Model derivation is *one of five* maximal decompositions, indexed by the Schläfli inventory of the cubic surface: 27 registers, 36 schedules, $40 + 40$ building parabolics, 45 polar pairs. The five vacua are connected by an explicit $5 \times 5$ double-coset transition matrix. Physically: what looks like "the" vacuum is a parabolic gauge choice, and the theory knows its own alternative phases and the exact combinatorial cost of moving between them. No continuum field theory has this property; here it is finite, exact, and enumerated.

**Matter is magic, with three octane grades.** The machine's non-classical resource (magic, in the resource-theoretic sense) coincides *exactly* with its matter sector: the 36 entangled Witting rays are the matter shell, and the 27 matter contexts stratify $[1, 8, 12, 6]$ — the cube's cell census. The three magic grades — deep $(2 + \sqrt{3})/6$ on $8 = 2^q$ rays, mid $(5 + 2\sqrt{3})/12$ on $24 = f$, shallow $3/4 = q/\mu$ (the Werner threshold) on $4 = \mu$ — mean that "what matter is" and "what fuels quantum advantage" are one census. A physicist reads this as a structural identification of the fermionic sector with the non-stabilizer sector; an engineer reads the same integers as a fuel gauge. Both readings are forced by the same double cosets.

**The Standard-Model spine: one transvection.** The preceding paragraphs assemble into a single statement, verified end-to-end in one pass (`bt886_standard_model_spine.py`): *from the pair* $(W(3,3),\ R)$ *— the symplectic quadrangle and one long-root transvection — the discrete Standard Model follows with zero free parameters.* The transvection $R$ fixes the 13-point gauge perp-plane and acts freely (nine orbits) on the 27 matter shell; its centralizer $C(R)$ (order 648) is the gauge group, with module $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \times SU(2) \times SU(3)$; its centre $Z(C(R)) = \langle R \rangle = \mathbb{Z}_3$ is the three generations; its normalizer gives the flavor group $S_3$ and charge-conjugation; the $W/Q$ duality gives parity; the matter shell grades 9+9+9 with the $\mathbb{Z}_3$ Yukawa rule; and the connection is flat ($\mathbb{Z}_3^2$) along collinear edges, $2T$-curved on the matter graph $Q$ with a quaternionic field strength. The gauge group, the generations, the flavor symmetry, the Yukawa texture, chirality, parity, and the gauge dynamics are not separate inputs — they are the centralizer, centre, normalizer, duality, grading, and commutator structure of one element of $\mathrm{Sp}(4, \mathbb{F}_3)$, replicated over the 40 points. The gauge group even carries its color/electroweak split: $C(R) = 3^{1+2} : SL(2,3) = 27{:}24$, with the normal Heisenberg radical (order $q^q = 27$, the color part) fixing the electroweak $\mathbf{1} \oplus \mathbf{3}$ pointwise (the $W/Z/\gamma$ are color-blind, constant on the four lines through $p_0$) and acting on the gluon octet $\mathbf{8}$, and the Levi $SL(2,3)$ (order $f = 24$, the electroweak part) carrying the $\mathbf{1} \oplus \mathbf{3}$ — color $\rtimes$ electroweak as a parabolic, with substrate orders $q^q$ and $f$. And the color radical is *the same* Heisenberg group $3^{1+2} = O_3$ that acts simply transitively on the 27 matter shell (the matter torsor): **color is the matter-shell Heisenberg group**, so matter carries color precisely because the 27-shell is a torsor under the color group — color rotations and matter-shell motions are one action of $q^q = 27$. Under the full gauge group the matter shell is the homogeneous space $27 = C(R)/SL(2,3)$ (gauge group over the electroweak Levi): $C(R)$ is transitive with point-stabiliser the electroweak $SL(2,3)$, rank 6 (suborbits $[1,1,1,8,8,8]$), module $\mathbb{C}[27] = \mathbf{1} \oplus \mathbf{6} \oplus \mathbf{8} \oplus \mathbf{12}$, and the three electroweak-fixed points form a color-singlet (lepton-like) 3-set while the three color-8 suborbits carry the colored (quark-like) content ($27 = 3 + 24$). One generation's fermion multiplets are the $C(R)$-module structure of the 27.

**Three generations, matter-blind and gauge-visible.** Because the matter register is the Steinberg module (Corollary, §10) and the Steinberg character vanishes on every 3-singular element, *any* order-3 symmetry of the substrate splits matter into exactly 27+27+27: the three generations carry identical matter/Steinberg structure — they are indistinguishable to the protected register, which is why generations share gauge charges. Yet the four order-3 conjugacy classes *are* distinguished in the gauge (40-point) sector: the transvections grade it $22 + 9 + 9$ and fix a whole $\mathrm{PG}(2,3)$ plane of $13 = \Phi_3$ points, while the regular classes grade it $16 + 12 + 12$ and fix only $\mu = 4$ points. The *physical* texture triality (Pillar 68's Yukawa map $R$) is identified exactly: it is the long-root transvection (a 40-class element = the centre of the point-parabolic's Heisenberg radical), fixing the 13-point gauge perp-plane $p_0^\perp$ pointwise and acting on the 27-point matter shell with 9 free orbits of 3 — the canonical "fix the gauge, stir the matter" generator. Its eigengrading of $\mathbb{C}[27]$ is $9 \oplus 9 \oplus 9$, and $\mathbb{Z}_3$ Clebsch–Gordan then *forces* the Pillar-68 Yukawa selection rule $T[a, b, v] = 0$ unless the three grades sum to 0 mod 3 (exactly 9 of 27 grade-triples survive); the matter coupling lives on the 36 within-shell tritangent triples (Pillar 69's Heisenberg-twisted triads, $45 = 9 + 36$). And the gauge side closes it: the centralizer $C(R) = \mathrm{Stab}(p_0)$ of order 648 (the gauge group is what commutes with generation-cycling) acts on the $12 = k$ gauge neighbours with permutation rank 3, decomposing the gauge module as

$$\mathbb{C}[12] = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \oplus SU(2) \oplus SU(3),$$

the Standard-Model gauge group adjoint (the **1**+**3** from the $A_4$ action on the four lines through $p_0$, the **8** the within-line gluon octet). So $R$ fixes the gauge sector ($12 = 8{+}3{+}1$, generation-blind) and grades the matter sector ($27 = 9{+}9{+}9$, three generations): the gauge group and the generations are the two eigen-data of one long-root transvection. The gauge sector even carries a parity statement: $C(R)$ acts on the four lines through $p_0$ as $A_4$ inside PSp, but as the full $S_4$ in $\mathrm{PGSp}(4, \mathbb{F}_3) = W(E_6)$ — the odd (parity-violating) 4-line permutations are *exactly* the anti-symplectic $W/Q$-duality coset, so gauge parity is the duality/chirality $\mathbb{Z}_2$ and lives only in the full $W(E_6)$. The flavor sector even has its charge conjugation: $N_{W(E_6)}(\langle R \rangle)/C(R) = \mathbb{Z}_2$, so there is an element $C$ with $CRC^{-1} = R^{-1}$ that fixes generation grade-0 and swaps grade-1 $\leftrightarrow$ grade-2 — the discrete shadow of CP, conjugating the generation $\mathbb{Z}_3$ exactly as the temporal Bell qutrit's CPT conjugates $\omega \mapsto \bar{\omega}$. The full Standard-Model discrete flavor/parity data — gauge group $1 \oplus 3 \oplus 8$, three generations $\mathbb{Z}_3$, generation $C$, matter chirality, gauge parity — is the normalizer geometry of one long-root transvection. The grading $R$ and its conjugation $C$ together generate the **flavor group** $S_3$ ($\langle R, C \rangle$, order 6, non-abelian — the minimal non-abelian flavor symmetry of BSM model-building), under which the matter shell decomposes as $\mathbb{C}[27] = 6 \cdot \mathbf{1} \oplus 3 \cdot \mathbf{1}' \oplus 9 \cdot \mathbf{2}$, the standard doublet appearing $q^2 = 9$ times. And the flavor–gauge relationship is exact: $R$ is central in the gauge group $C(R)$, with $Z(C(R)) = \langle R \rangle \cong \mathbb{Z}_3$ — **the three generations are the centre of the gauge group**, acting trivially on the adjoint exactly as the $\mathbb{Z}_3$ centre of colour $SU(3)$ acts on the gluon octet (so the $SU(3)$ centre *is* the generation $\mathbb{Z}_3$). Globally there are 40 such local gauge groups — one per point, a single conjugacy class (homogeneous spacetime) — and the 40 generation-centred long-root $\mathbb{Z}_3$'s generate all of $\mathrm{PSp}(4, \mathbb{F}_3)$: **the** 40 **points of** $W(3,3)$ **are the space of local gauge groups**, each $SU(3) \times SU(2) \times U(1)$ with its generation centre, and the global symmetry is the closure of the local generation symmetries. The connection between adjacent local gauge groups is exact: for collinear points the generation centres commute ($\langle R_p, R_{p'} \rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$), so the gauge connection is **flat along the** 240 **edges**; for non-collinear points $\langle R_p, R_{p'} \rangle = SL(2,3) = 2T$ (the binary tetrahedral / 24-cell group), so it is **curved across the** 27**-per-point matter shell**. Gauge curvature lives exactly on the matter graph $Q$ — flat = collinearity, curved = non-collinearity = matter, with holonomy the 24-cell group. The curvature 2-form $F(p,q) = [R_p, R_q]$ makes this quantitative: it vanishes on collinear (causal) directions and is an *order-*4 *quaternion unit* of $2T$ (one of $\pm i, \pm j, \pm k$, with $F^2 = -I$ the centre) on every matter pair — an $\mathbf{su}(2)$**-valued field strength supported exactly on the matter graph** $Q$. Flat causality, quaternionic curvature on matter. The integrated flux (Wilson loop $R_a R_b R_c$) confirms it: all 160 collinear triangles carry order-3 abelian flux (the flat $\mathbb{Z}_3$ line grading), while the 3240 matter triangles of $Q$ carry non-abelian flux of orders $\{2, 4, 6, 12\}$ (counts $180, 180, 1440, 1440$) up to $12 = k$ — gauge energy lives on the matter graph. The matter register (Steinberg) couples to this flux only through its non-3-singular part: by Steinberg vanishing, $\chi_{\mathrm{St}}(W) = 0$ on every order-$3, 6, 12$ loop (3040 triangles, matter-invisible) and is nonzero only on the order-2 loops ($\chi_{\mathrm{St}} = +q^2 = 9$, the polar-pair chirality involution of BT869) and order-4 loops ($\chi_{\mathrm{St}} = -q = -3$); the total coupling $\sum \chi_{\mathrm{St}}(W) = 180 \cdot 9 - 180 \cdot 3 = 1080 = 540 \cdot 2$ is exactly the chart double-cover count — gauge dynamics and the routing fabric meet at 1080. This is the Standard-Model pattern as a representation- theoretic theorem: generations are identical in gauge charge (matter-blind) and differ only through gauge-sector–visible couplings (the Yukawa/mass structure), here the eigenvalue degeneracy of an order-3 symmetry rather than a modeling input. *Witnesses:* `bt863_generations_from_steinberg_vanishing.py`, `bt864_triality_census.py`. Order-6 symmetries see *both* internal gradings at once: an order-6 element factors as $g^2$ (generation, order 3) times $g^3$ (chirality, order 2 — the sign-twisted carrier of $H_2$), and since $\chi_{\mathrm{St}}$ vanishes on every 3-singular power, its $\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$ eigengrading factors cleanly: each chirality half $(81 \pm \chi(g^3))/2$ carries 3 equal generations. For the dominant involution type ($\chi(g^3) = q^2$) the chirality split is $\mathbf{45} + \mathbf{36}$ — the substrate's own tritangent and double-six counts — so the left/right matter split is tied to Schläfli geometry; the other type gives $39 + 42$. Generation $\times$ chirality $= 3 \times 2 = 6 = q!$. The dominant chirality involution is identified exactly: $\mathrm{PSp}(4, \mathbb{F}_3)$ has just two involution classes, and the 45-class (eight fixed points carrying the cube $Q_3$, centralizer $2T \times 2T$ of order 576) is the *central involution of a hyperbolic polar pair* $L \leftrightarrow L^{\perp}$ — so the left/right split of matter, $45 + 36$, is sized by the substrate's own tritangent-plane and double-six counts, and the matter chirality $\mathbb{Z}_2$ *is* the 24-cell polar-pair swap. *Witnesses:* `bt868_joint_generation_chirality_grading.py`, `bt869_involution_chirality_classes.py`.

**Time has three hands and an irrational escapement.** The internal $\mathbb{Z}_{12}$ rectangle clock, the external $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) references, and the Boerdijk–Coxeter drive $\arccos(-2/3)$ together realize a discrete time quasicrystal: a drive that never repeats yet has exactly bounded gap structure (two gap lengths at $n = 30 = h(E_8)$). The arrow of time appears as the unique self-loop fixed point $I/q$ with its two-trit forgetting cost: irreversibility is priced, not postulated. Implication: any laboratory system realizing this architecture is also a clock whose long-term stability is protected by a number-theoretic irrationality rather than by feedback.

**The exceptional polytopes are the theory's siblings, not its parts.** The Grünbaum–Coxeter arc (Theorems 12.2–12.4) shows the substrate carries the complete local geometry of the 11-cell and 57-cell — hemi-icosahedra and hemi-dodecahedra in every schedule, with exact incidence — while universality plus Lagrange forbid their global crystallization inside $\mathrm{Sp}(4, \mathbb{F}_3)$. The same local amalgam $A_5 *_{D_{10}} A_5$ closes as $L_2(9)$ (the substrate's schedule core), $L_2(11)$ (the 11-cell), $L_2(19)$ (the 57-cell), with $J_1 \times L_2(19)$ — a sporadic group — atop the universal ladder. The physical moral is delocalization: the substrate *cannot* host a "GC condensate," so the icosahedral order is necessarily spread over the 36 schedules as compass structure. This is the same mathematical mechanism by which frustrated phases in condensed matter refuse local order parameters — realized here exactly, with the obstruction an integer $(11, 19 \nmid 25920)$.

**Spectral ghosts.** The primes that refuse to divide the group still govern its analysis: $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$, 19 enters through the Heawood genus $H(19) = 20 =$ the BC ring count, and the chart-web eigenvalue fields are $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ ($\Theta = 10$, $\Phi_{12} = 73$, both parameter-closure constants). A spectroscopist's summary: the machine's transport spectrum speaks the tier-ladder dialect of the physics paper, and the forbidden primes appear as poles of the zeta function rather than as symmetries.

**Gravity counts the matter register.** The discrete gravitational bridge is the Matrix-Tree action $S = \frac{M_P^2}{2} \ln \tau(G)$, $\tau$ the spanning-tree count, and for the substrate it is exact: the Laplacian spectrum $\{0, 10^{24}, 16^{15}\}$ gives

$$\tau(W(3,3)) = \frac{10^{24} \cdot 16^{15}}{40} = 2^{81} \cdot 5^{23},$$

where the exponent of 2 is $81 = q^4 =$ the matter-sector (Steinberg) dimension, the base $5 = F_5$ is the tier-ladder prime ($r = 27/80$), and the exponent $23 = f - 1$. The complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ gives $\tau(Q) = 2^{66} \cdot 3^{39} \cdot 5^{23}$ with 3-exponent $39 = q\Phi_3 =$ the gauge-sector dimension. So the discrete-gravity partition function carries the Standard-Model sector sizes as its prime-exponent charges — matter ($q^4$) in the power of 2, gauge ($q\Phi_3$) in the power of 3, cosmological ($F_5$) shared — and the entropy of the spanning-tree ensemble *counts the registers.* This is not a separate fact from the transport spectrum: the Ihara zeta $\zeta^{-1}(u) = (1-u^2)^{200}(1-u)(1-11u)(1-2u+11u^2)^{24}(1+4u+11u^2)^{15}$ (verified directly by the $480 \times 480$ Hashimoto operator) has its non-trivial zeros on the graph-RH circle $|u| = 1/\sqrt{11}$ (the transport sector, $p_{\mathrm{Ih}} = k-1 = 11$) and its Perron zero at $u = 1$, where the Bass matrix $I - A + 11I = 12I - A$ *is* the Laplacian, so the Matrix-Tree leading coefficient $10^{24}16^{15} = 2^{84}5^{24} = v \cdot \tau$ recovers exactly the gravity count. Transport and gravity are the off-critical and critical behavior of one zeta: the prime 11 is the transport critical norm, the matter dimension $q^4 = 81$ is the gravity entropy. The substrate is in fact a *pair* of zetas: the complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ has its own Ihara prime $26 = 2\Phi_3$ (Hashimoto Perron, all complex eigenvalues on $|u| = 1/\sqrt{26}$) and its Perron-point gravity $\tau(Q) = 2^{66}3^{39}5^{23}$ puts the *gauge* dimension $39 = q\Phi_3$ where $W(3,3)$ puts the matter dimension — complementation (collinearity $\leftrightarrow$ non-collinearity) swaps the matter and gauge entropies, with the cosmological charge $5^{23}$ shared. The non-backtracking walk census confirms the dynamics: $N_3 = \mu|E| = 960$, $N_5 = |E|\, q^q\, n_{\mathrm{even}} = 181440$, $N_n \sim 11^n$. *Witnesses:* `bt870_spanning_tree_gravity.py`, `bt872_ihara_zeta_spanning_bridge.py`, `bt873_dual_zeta_walk_census.py`.

**Falsifiability is retail, not wholesale.** Each physics reading above is pinned to a tabletop witness with an exact rational value: trace–Choi visibility $1/3 = 1/q$ exactly (not approximately); Kochen–Specker budget $36/40 = (q!)^2/v$ exactly, with deficit $\mu$; contextual fraction $1/10 = 1/\Phi_4$; Werner threshold $3/4 = q/\mu$; two gap lengths — not three — at drive step 30. A single wrong integer kills the corresponding layer. This is a stronger falsifiability posture than parameter-fitting physics can offer, because there are no parameters to refit.

## 14.2 Computing

**What the machine is.** One photon carries a 4-dimensional path–polarization register (states) and a 9-dimensional past–future time-bin register (operators), realizing the same $W(3,3)$ geometry twice; the operator–operand duality makes program and data literally the same incidence object. Universality is a two-ingredient theorem: three passive optical elements (tritter, phase plate, EOM) generate the full 51840-element Clifford symplectic closure — *exactly*, not approximately — and the matter shell supplies the magic states that promote Clifford to universal. There is no cryostat, no vacuum chamber, no trapped ion: the exotic resource is geometry.

**What it costs.** The build sheet is a heralded single-photon source, one polarizing beam splitter, one symmetric tritter, a three-rail delay ladder, one electro-optic modulator, one recirculation loop with a fixed irrational rotation, and single-photon detectors. Every component is catalog optics. The machine's benchmarks are integer-valued: a laboratory can certify the Clifford layer by counting to 51840 and certify the fuel by hitting 36/40. Self-testing is native because the specification is arithmetic.

**The software stack exists and is finite.** Programs are braid words (with $\sigma^5 = Z$ exact); the instruction set is the BT828 compiler ABI; the operating system is the timetable library (36 schedules; overlap law 1-or-4 with switching cost 6); synchronization is the beacon-heptad mesh (2880 frames, themselves forming an exchange network); memory protection is the Steinberg sector ($81 = q^4$, a single irreducible — protected by Schur's lemma, the strongest no-leakage statement representation theory can make). The state/program compiler makes that register directly addressable: $\mathbb{F}_3^3$ supplies three commuting copies of the 27-slot program shell, $H_{27}$ supplies three noncommuting copies of the state shell, and its center supplies the native generation flag $27 \subset 54 \subset 81$. The oriented timetable header above that memory splits as $5_\omega + 5_{\omega^2} + 30$: the full Weyl runtime fuses the handed pair into a 10-channel and leaves one 30-channel with an outer parity choice. Cache lookup is a distinct split layer: two copies of $H/C_4$ share an 81-route $H/D_8$ base, and their 324 slots form 81 four-state $D_8$ address fibers. The non-split ternary transport operator then updates the selected state; address choice and temporal memory cannot alias because their minimal polynomials differ. The watchdog is the BC loop. Each layer is a theorem; the stack compiles because double cosets compose.

**Redundancy is built in, not bolted on.** The newest results sharpen the fault story: each pentad core stores its timetable *three independent ways* (lit-chart transversals, the $K_5$ edge labeling, dark-sector shadows), every skew pair has six Petersen homes and two matching covers, and the dark sector carries its own chart matching and two chiral tetrad partitions (Theorem 12.4 block). Erasure of any single description leaves the schedule reconstructible — redundancy appears as a counting identity, not as an engineering allowance. The middleware's two packet phases (tetrahedral/hemioctahedral, the tomotope's $2_{\{0,1,2\}}$ class) confine phase changes to cell hops, which is where the commit boundaries already sit, and the runtime's vertex layer is a free $\mathbb{F}_2^2$-torsor (the Cayley structure), so address arithmetic is XOR end to end.

**What it is not.** No claim is made here of asymptotic speedups over error-corrected gate-model machines, of fault-tolerance thresholds, or of scalability beyond the fractal substitution rule ($40^n$ leaves, diameter $8n$, commit clock $T(n) = 4(7^n - 1)$) — those are stated as exact combinatorial scaling laws, and their physical realization at depth $n > 2$ is an open engineering problem with its first hard boundary already computed (the $n = 5$ desynchronization with remainder $24 = f$). The honest claim is narrower and stranger: a complete, finite, self-verifying computational architecture exists inside one photon's worth of quantum optics, with zero adjustable parameters.

### 14.3 Networking

**The network is the computer, twice over.** The thesis is not rhetorical: the atlas's transport groupoid and the register's gate groupoid are the same groupoid, so moving a packet and applying a gate are one operation. The new library-geometry theorem (Theorem 12.2) doubles the statement at the control plane: the 36 timetables form SRG$(36, 15, 6, 6)$, the *other* rank-3 geometry of the same group, so the control plane and the data plane (SRG$(40, 12, 2, 4)$) are the two classical geometries of one symmetry. Configuration management inherits strongly-regular guarantees: any two timetables, near or far, have exactly six common near-neighbors — worst-case reconfiguration paths are bounded by the geometry itself.

**Routing, addressing, switching.** Local routing is e-cube XOR on $Q_3$ charts (540 of them; the atlas is glued along the 1620 apartments of the Tits building, diameter 5). Addresses are Gray-coded $\mathbb{F}_2^3$ words; the antipode map is bit complement. Timetable switching obeys the overlap law: cheap hops preserve $4 = \mu$ of 10 calibrated contexts, the switch graph is the $U_4(2)$ rank-3 geometry, and every skew pair lies in exactly 3 schedules, 6 Petersen homes, and 2 pentad-matching covers — the router's spare-path inventory is an exact census, not a provisioning estimate.

**Synchronization without a master.** The beacon heptads (2880 complete $K_7$ interference meshes, all pairwise overlaps 1/3) are reference frames that form their own exchange network — the network's clock distribution is itself a network, with no distinguished master node. External discipline comes from the $\mathbb{Z}_7/\mathbb{Z}_{13}$ clocks (both decimal period 6) and the BC quasicrystal drive; internal phase order from $C_{12}$. A distributed-systems reader will recognize the structure: consensus by geometry, where the "leader election" problem dissolves because the frame family is a single transitive orbit.

**Security posture (implication, flagged).** Three structural facts have security readings, stated here as directions rather than theorems. (i) Contextuality is tamper-evidence: the KS budget 36/40 is exact, so any channel intervention that classicalizes measurement statistics moves an integer and is detectable in principle. (ii) Gate teleportation stores a unitary in the photon's self-entanglement and applies it by Bell projection against its own past — a store-and-forward primitive for quantum repeaters that never exposes the gate classically. (iii) The timetable redundancy (three independent reconstructions) is an erasure-code posture against denial: losing a description class does not lose the schedule. Turning these into security proofs requires adversary models this paper does not supply.

**Scaling.** The fractal holonet substitutes a full 40-node substrate for each leaf: $40^n$ leaves, $(40^n - 1)/39$ substrate instances, reversible diameter $\leq 8n = 8\log_{40} N$, durable commit clock $T(n) = 4(7^n - 1)$, with the first guard band an arithmetic necessity at $n = 5$. Because every child inherits the same finite protocol, scaling adds no new control logic — the protocol stack is depth-invariant. This is the architectural property that datacenter fabrics approximate statistically and this geometry has exactly.

### 14.4 The physics-to-architecture dictionary

The dictionary is the deepest implication: one set of integers reads simultaneously as physics and as engineering, because both readings are projections of the same incidence theorems.

| object | physics reading | architecture reading |
|---|---|---|
| $1 + 12 + 27$ split | Bell line + meeting + disjoint | pole / I-O ports / fuel+store |
| 240 edges | interaction qutrits | bandwidth = memory |
| $81 = q^4$ Steinberg | protected sector (Schur) | logical register |
| dual $O_3$ restrictions | Heisenberg state phases | state/program compiler |
| $H_2 = 5 + \bar{5} + 30$ | oriented context chirality | timetable header $10 + 30^{\pm}$ |
| $324 = 81 \times 4$, commutant $D_8$ | split cache multiplicity | four-state address fiber |
| $0 \to 81 \to 162 \to 81 \to 0$ | ternary temporal extension | non-split state update |
| 36 spreads | measurement bases | timetable OS |
| $\mathrm{SRG}(36, 15, 6, 6)$ | basis-overlap law | control-plane fabric |
| five vacua | phases of the theory | boot modes |
| magic strata 8+24+4 | fermion shell grading | fuel octanes |
| 540 skew pairs | two-fold null systems | hypercube charts |
| 1620 apartments | Weyl frames | inter-chart trunks |
| 2880 heptads | equiangular frames | sync beacons |
| BC drive $\arccos(-2/3)$ | quasicrystal time | watchdog clock |
| compasses (216+216) | icosahedral order, delocalized | needle registers |
| pentads (chiral 216+216) | maximal partial spreads | $\mathrm{F}_5$ chart caches |
| dark sector zoo | hidden-half structure | erasure store |
| tomotope ($2_{\{0,1,2\}}$, Cayley/$V_4$) | sphere/projective alternation | packet middleware |
| 11-cell, 57-cell, $J_1$ | universal completions (blocked) | the covers the hardware refuses |

The first row carries the deepest identification, drawn from the companion paper *Universal Quantum Computation Through a Single Photon*: the split $1+12+27$ is not merely the local SRG anatomy of a point — it is the **Bell-line shell** of the photon's own self-entanglement seed. The temporal Bell qutrit $|\Omega\rangle$ stabilizes one line of $W(3,3)$; the 40 lines then decompose as 1 (the Bell line itself) $+12$ (lines meeting it — the gauge sector) $+27$ (lines disjoint from it — the matter sector), with substrate reading $1+k+q^q$. The vacuum decomposition that organizes the Standard-Model derivation is literally the photon's view of its own now-context: *self = the seed of self-entanglement*. The Bell-line stabilizer has order $51840/40 = 1296 = \mu^2 q^{q+1}$ — the same $1296 = 6^4$ that counts the compass $D_{10}$-incidence pairs of Theorem 12.4 (36 schedules $\times$ 36 core pairs), but Theorem 2.3 proves that this is not a regular-action identification. Instead, the projective Bell stabilizer resolves the incidence carrier as $648+324+162+162$, the exact mirror/schedule/left-cache/right-cache prefix decoder, while the point parabolic supplies two persistent 648 address sheets. The two cache words 110 and 111 are now resolved more finely by Theorem 10.6: they name two copies of the same $H/C_4$ cache, not two inequivalent internal chiral modules. Both double-cover one canonical $H/D_8$ base of 81 routes, and the complete cache quarter is an exact $D_8$-over-81 square bundle. This reversible address plane is provably not the non-split $81 \to 162 \to 81$ transport plane, despite their common 162 count. The companion also supplies the master equation at Hilbert-space level ($q^2 = q + q!$: the 9-dimensional past$\otimes$future space splits into 3 diagonal "now" histories and $6 = q!$ context-changing ones), and a ledger identity: temporal gate teleportation consumes exactly 2 classical trits of memory — the same 2-trit cost that prices the arrow of time in the self-loop fixed point of §11. Storing a state for one's future self and forgetting one's past cost the same two trits. The 27 itself now has a dual decoding: the 27 non-collinear *points* form a torsor under the genuine Heisenberg group (the normal $3_+^{1+2}$ of the point parabolic, acting simply transitively), whose invariant relations reconstruct the whole classical 27-zoo — the 9-triangle fibration, $\mathrm{GQ}(2,4) = \mathrm{SRG}(27,10,1,5)$, and the Schläfli graph $\mathrm{SRG}(27,16,10,8)$ of the $E_6$ 27-lines world — while the 27 disjoint *lines* of the Bell shell form a torsor under elementary $\mathbb{F}_3^3$ (the line parabolic's $O_3$), a flat 3-trit register cube carrying *no* invariant strongly regular graph. Curvature for states, flatness for programs: the matter sector is a Heisenberg manifold on the state side and a ternary address space on the operational side. The register reading is exact: assigning each Bell-shell context its 3-trit address, the four pair relations $[4,4,6,12]$ are the difference-vector classes, and the geometry is a function of register arithmetic alone — meeting contexts ($\pm$-paired 4-classes) share exactly 1 transversal onto the Bell line, near-skew (12) exactly 2, far-skew (6) exactly 0: base-3 register incidence for matter contexts, twin to the charts' base-2 Gray arithmetic. *Witnesses:* `bt858_heisenberg_shell_torsors.py`, `bt860_bell_shell_register_arithmetic.py`.

Three rows deserve emphasis. The *Steinberg row* is the strongest protection statement available: the register is one irreducible representation, so *any* equivariant leakage operator is zero by Schur's lemma — protection by symmetry, not by redundancy. The *compass rows* say that what physics sees as delocalized icosahedral order, engineering sees as a navigation instrument: 12 needles per schedule, each carrying Petersen addressing, five-chart caches, and a dark-sector mirror. The *universal-polytope row* states the program's sharpest structural lesson: the hardware keeps the local physics of the exceptional objects and refuses their global rigidity; the refusal (an integer non-divisibility) is precisely why the architecture is distributed.

### 14.5 Beyond the laboratory

**A different contract between theory and experiment.** Every layer of this machine is specified by integers and verified by scripts that re-derive those integers from first principles in seconds. The cultural implication is a publishing model in which papers *run*: a claim is not a sentence but a reproducible computation, and a correction is a commit. This program has already operated under that contract (see the ethos below) — twice it corrected its own published claims at the source, and the corrected values were more natural than the originals. If the broader physics literature adopted executable claims at even a small scale, entire classes of irreproducibility would become type errors.

**Quantum hardware without a building.** Architectures that need millikelvin cryostats centralize by economics: only institutions can own them. A universal architecture whose parts list is catalog optics — source, splitters, delay fiber, modulator, detectors — decentralizes by the same economics. The realistic near-term role is not supremacy-class computation but *certified* small-scale quantum information: teaching laboratories that can verify 51840 and 36/40 on a bench; metrology nodes whose clock is protected by an irrational rotation; network testbeds where gates and routes are the same hardware. A machine whose benchmarks are exact integers is also the natural pedagogical machine: a student can check the whole specification.

**Communication and computation stop being separate utilities.** In this architecture there is no boundary at which computation hands off to networking — the gate groupoid and the transport groupoid coincide, and the fractal substitution makes the computer/network identity scale-invariant. The long-run implication, if any physical system realizes this pattern at depth, is infrastructural: bandwidth and compute become one provisioned quantity, scheduling becomes group theory, and the spectrum-allocation problem for the network's measurement contexts is solved by a finite library (36 timetables with a known switching metric) rather than by auction or contention.

**Society-scale stakes of a zero-parameter physics.** Separately from the machine, the substrate program's physics claims — 54+ observables, 14 falsifiable predictions, no adjustable parameters — carry a specific societal meaning: they are maximally exposed. A theory with no dials cannot retreat; the 2027 neutrino-hierarchy determination, the proton-decay window, the CMB measurements will each either hit the predicted integers or end the corresponding pillar. Public trust in fundamental science is built by exactly this posture — predictions that can die — and the machine half of the program extends the posture to hardware: the device specification is its own audit.

**What could go wrong, said plainly.** The architecture is exact; its physical realization is not yet demonstrated beyond the component level. The fractal scaling laws are combinatorial theorems, not engineering schedules. The security readings are directions, not proofs. And the identification of the substrate with fundamental physics stands or falls with the prediction table, not with the elegance of the mathematics. This paragraph exists because the program's ethos (below) requires the failure modes to be listed next to the claims.

### 14.6 The ethos

Several pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3,3)$ is 7, not 10 (no ovoid exists; the "perfect graph" block is withdrawn); the Kochen–Specker optimum is 36/40, not 34/40; the pentad-sharing count of BT844 was corrected by the chiral-orbit census of BT845 within one working session; and the tomotope's framing was corrected from "non-polytopal maniplex" to "uniform polytope with non-polytopal minimal covers" when the literature was read precisely; and the "$\mathbb{Z}_3$ Berry phase as a discrete second Chern class" was corrected (BT871) — the uniform Berry 2-cochain is a coboundary over $\mathbb{F}_3$, so the physical $\mathbb{Z}_3$ is the order-3 action on the Steinberg register, not a curvature class.

In every case the corrected value is *more* substrate-natural than the claim it replaced ($\alpha = \Phi_6$; budget $= (q!)^2/v$; $432 = 216 + 216$ chiral; infinitely many covers as the actual pathology; the $\mathbb{Z}_3$ as a group action rather than a cocycle) — the pattern a real structure shows when probed honestly. The program treats corrections as first-class results: they are committed, logged, and cited like theorems, because a corpus that cannot heal is a corpus that cannot be trusted.

# A Verification ledger

Every theorem above is backed by an executable script in `analysis/` with results in `data/`; the chain BT739–BT839 was developed and pushed incrementally with exact arithmetic (GF($p$) elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

| layer | primary witnesses |
| --- | --- |
| duality / carriers | bt817, bt820, bt821 |
| atlas / building / web | bt773, bt777, bt744, bt779 |
| mirror middleware / tomotope runtime | bt814, bt815, bt826 |
| braids / registers | bt740, bt741 |
| vacua / geography / transitions | bt808–bt813, bt816 |
| fuel / contextuality | bt818, bt822, bt823 |
| memory / Steinberg | bt742, bt744 |
| clocks / quasicrystal / arrow | bt774, bt819, bt820, bt823 |
| sync / schedules / overlap law | bt817, bt819, bt835 |
| Grünbaum–Coxeter hemicells / library geometry | bt836, bt837, bt839 |
| two compasses / $4 \cdot K_{10} = 12$ Petersens / chart double cover | bt843–bt845 |
| pentad anatomy / schedule reconstruction / dark dodecahedra | bt846, bt847 |
| amalgamation ladder / universality / $J_1$ | bt848 |
| maniplex barren rung / class $2_{\{0,1,2\}}$ / Cayley over $V_4$ | bt849–bt851 |
| seventeen universals / Aut(tomotope) universal | bt852 |
| dark orbit zoo / shadow bijection / $K_5$-complete dark sector | bt853–bt855 |
| dark charts = mirror bus ($G$-set iso) | bt856 |
| Heisenberg/$\mathbb{F}_3^3$ shell torsors, Schläfli decoding | bt858 |
| Bell-shell register arithmetic ($[4,4,6,12]$ named) | bt860 |
| code register = Steinberg ($[[240,81,4,3]]_3$) | bt861 |
| $H_2$ = sign-twisted lines (homological dictionary) | bt862 |
| three generations = Steinberg vanishing | bt863 |
| triality census (matter-blind, gauge-visible) | bt864 |
| joint generation $\times$ chirality grading | bt868 |
| chirality = polar-pair involution $(45+36)$ | bt869 |
| spanning-tree gravity $\tau = 2^{81}5^{23}$ | bt870 |
| Berry-phase correction ($\mathbb{Z}_3$ is the Steinberg action) | bt871 |
| Ihara zeta = spanning-tree gravity bridge | bt872 |
| two-zeta duality / walk census | bt873 |
| texture triality = long-root transvection | bt874 |
| Yukawa rule = $\mathbb{Z}_3$ grade conservation | bt875 |
| gauge group $1{\oplus}3{\oplus}8$ from $C(R)$ | bt876 |
| gauge parity = duality ($A_4{\to}S_4$) | bt877 |
| generation charge-conjugation ($N/C = \mathbb{Z}_2$) | bt878 |
| flavor group $S_3$ ($\mathbb{C}[27] = 6{\cdot}1{+}3{\cdot}1'{+}9{\cdot}2$) | bt879 |
| generations = centre of the gauge group | bt880 |
| spacetime = space of 40 local gauge groups | bt881 |
| gauge connection flat on edges, $2T$-curved on $Q$ | bt882 |
| curvature 2-form (quaternionic, on $Q$) | bt883 |
| gauge flux (Wilson loops, $\mathbb{Z}_3$/lines, $\leq 12/Q$) | bt884 |
| YM–matter coupling $\sum \chi_{\mathrm{St}} = 1080$ | bt885 |
| SM spine (one transvection, master theorem) | bt886 |
| gauge group = color : electroweak (27:24) | bt887 |
| color = the matter-shell Heisenberg group | bt888 |
| fermion content $27 = C(R)/SL(2,3)$ (lepton/quark) | bt889 |
| dual-torsor Steinberg compiler / native generation flag | bt865 |
| oriented $H_2$ irreducible spectrum / outer lift | bt866 |
| cache $D_8$-over-81 fibration / split–non-split boundary | bt867 |
| chirality bookkeeping / Bell–compass parabolic router | bt857, bt859 |
| universality | bt825 |
| fractal computer/network | bt827 |
| runtime compiler / sentinel / commit / cover boundary | bt828–bt834, bt838 |
| corpus corrections | bt818, bt823, bt824 |

## B Architecture Completeness and the Three Physical Residuals

The holonet machine is now specified end to end: the 40-point substrate register and its $W(3,3)$ commuting-projector Hamiltonian; the $[[240,81,4]]_3$ Steinberg code as the native error-correcting register; the local gate set and transversal/holonomy operations; the gauge connection (flat along the 240 edges, curved on the matter graph with binary-tetrahedral $2T$ holonomy) as the physical wiring of multi-qubit interaction; and the discrete dynamics, in which the physical on-shell states are exactly the harmonic forms $\ker D$ (vacuum $\oplus$ Steinberg matter $\oplus$ line module $= 1 \oplus 81 \oplus 40$) — the least-action ground space of the finite spectral action $\operatorname{Tr} f(D^2)$, whose value reproduces the spanning-tree gravity invariant. *As a universal computer the architecture is complete*: universality, the code, the gate set, and the runtime compiler are established (cf. the substrate ledger above).

What remains is not computational but *physical*, and the three residuals of the master manuscript's completeness boundary (`w33_paper.tex`) are now each resolved or reduced:

- **(R1) the gauge-lattice embedding — resolved.** The machine's symmetry register carries the canonical symplectic $E_8/2E_8$ datum mod 2, and that datum admits a *unique* $\mathrm{PSp}(4,3)$-invariant quadratic refinement, of plus type; hence $\mathrm{PSp}(4,3) \subset O_8^+(2) = \mathrm{Aut}(E_8/2E_8)$, and the canonical positive-definite lift is $E_8$ (up to a central $\{\pm 1\}$). The gauge lattice is fixed.
- **(R2) the symmetry / matter register — datum supplied, bridge built.** The quintic moonshine traces $\mathrm{Tr}(2A|V_5) = 74428120$, $\mathrm{Tr}(2B|V_5) = 184024$ are computed; and the matter register is realized as a finite spectral triple (a Hilbert–Schmidt bimodule of dimension 81, first-order condition verified, gauge content $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$). The residual is now the explicit physical fermion-multiplet assignment, a representation- selection task.
- **(R3) the emergent-spacetime limit — theorem-governed.** On a shape-regular (edge-wise) refinement tower the full gravity-plus-matter spectral action converges term by term to the continuum Einstein–Hilbert + Standard-Model action — by Cheeger–Müller–Schrader (curvature), higher-order Regge (higher-derivative), classical lattice gauge, and the exact finite moments $\{440, 1920, 16320\}$; the spectral action is moreover continuous for Latrémolière's spectral propinquity. The emergent-spacetime limit is thus the *application* of established convergence theorems, not an open analytic obstruction.

None of the three is a gap in the machine's *operation* — the holonet runs, corrects, and computes universally as specified — and none is now an open problem in principle: a fixed gauge lattice ($E_8$), a matter register awaiting only its physical labelling, and an emergent spacetime with a theorem-governed continuum limit. This is the precise sense in which the architecture is complete: a finite, exact, universal machine whose physics closes onto the Standard Model and general relativity through results that are now either proved or reduced to the application of known theorems.

### B.1 The architecture is the substrate's symplectic continuum

There is a structural reason the machine and the spacetime are different objects built from the same $W(3,3)$: the substrate has *two* continuum limits. The arithmetic tower $W(3,q)$ (the symplectic quadrangle over $\mathbb{F}_q$, $q = 3^n$) is spectrally rigid — exactly three eigenvalues at every $q$, no Weyl law — so it never becomes a smooth spacetime; the metric continuum needs the external $K3$ seed (the physics side, `w33_paper.tex`, Prop. on the two continua). The *intrinsic* continuum of $W(3,3)$ is instead *symplectic*: its Heisenberg matter shell $3^{1+2}$ carrying the $\mathrm{Sp}(4,3)$ action is a Weil-representation datum, whose $q \to \infty$ limit is the metaplectic (oscillator) representation of $\mathrm{Sp}(4,\mathbb{R})$ on $L^2(\mathbb{R}^2)$ — precisely the continuous-variable, Fock/oscillator computation this paper realizes. So the holonet *is* the symplectic continuum of the substrate, exactly as the spacetime is its metric continuum: the universal computer and the universe are the two faces of one finite geometry's continuum limit. (Witness: `analysis/w33_two_continua_symplectic_metric.py`.)

### B.2 The fault-tolerant layer is the substrate's lattice tower

Because the holonet's logical layer is a continuous-variable (CV) oscillator computation, its canonical fault-tolerant encoding is the Gottesman–Kitaev–Preskill (GKP) code, which embeds a qudit in an oscillator. A *multimode* GKP code *is* a symplectic lattice in phase space: stabilisers are displacements by lattice vectors, and error correction is closest-point decoding in the symplectic dual lattice. The code quality is set entirely by the lattice, and the established optima are exceptional lattices: the *hexagonal* lattice $A_2$ for one mode, the densest 4-dimensional lattice $D_4$ for two modes (strictly better than any product of one-mode codes), and $E_8$ for four modes [20, 21].

This is not a free design choice for the holonet: the substrate *fixes* the lattice at every scale. The machine is two qutrits, hence two oscillator modes ($L^2(\mathbb{R}^2)$, phase space $\mathbb{R}^4$), so its optimal GKP code lattice is $D_4$ — and $D_4$ is exactly the substrate's matter shell, with $|W(D_4)| = 192$ the tomotope/flag symmetry and $D_4$ triality the three-generation structure. The single-mode optimum $A_2$ is the $q = 3$ (hexagonal, $SU(3)$) lattice of a single qutrit, and two modes stacked, $D_4 \oplus D_4$, sit primitively inside $E_8$ — the substrate's mod-2 homology lift (R1), which is also the $K3$ gauge lattice. Thus the tower

$$A_2 \subset D_4 \subset E_8$$

is *simultaneously* the optimal 1-, 2-, 4-mode GKP code lattices (the architecture's quantum error-correction layer) and $W(3,3)$'s physical lattice tower (single-qutrit $SU(3)$, matter shell, gauge $E_8$). The error correction of the computer and the gauge structure of the universe are one and the same lattice tower; the QEC code is not chosen, it is the substrate. (Witness: `analysis/w33_gkp_lattice_architecture.py`, verifying the kissing numbers $6, 24, 240$ and the embedding $D_4 \oplus D_4 \subset E_8$.)

This also fixes the lattice part of the fault-tolerance threshold. The substrate lattices are *isodual* (symplectically self-dual — the balanced-GKP condition, $\hat{q}$ and $\hat{p}$ protected equally) and densest in their dimensions, so they maximise the nominal coding gain $\gamma = d_{\min}^2/\det^{1/n}$ among symplectic lattices of that rank:

$$\gamma_{\mathrm{dB}}(A_2) = 0.6, \qquad \gamma_{\mathrm{dB}}(D_4) = 1.5, \qquad \gamma_{\mathrm{dB}}(E_8) = 3.0$$

(and 6.0 for the Leech lattice). So the holonet's two-mode code $D_4$ already buys $\sim 1.5$ dB, and the four-mode code $E_8$ $\sim 3$ dB, of squeezing-margin over the trivial square code, for free. The absolute threshold still needs the noise model, the finite-squeezing GKP-state quality and the syndrome-extraction protocol — which we do not fix here — but the lattice contribution, the part that depends only on the code, is the substrate's and is optimal. (Witness: `analysis/w33_gkp_coding_gain.py`.)

### B.3 The universal gate set is degree two plus degree three

The code fixes the fault-tolerant layer; the gate set completes the architecture, and it too is substrate-forced. The continuous-variable universality theorem [22] states that the Gaussian unitaries — those generated by Hamiltonians at most *quadratic* (degree 2) in the mode operators: beamsplitters, phase shifters, squeezers, i.e. the CV Clifford group $\mathrm{Sp}(2n, \mathbb{R})$ — are *not* universal, but adjoining any single generator of degree $\geq 3$ (canonically the cubic phase gate $e^{i\gamma\hat{q}^3}$) generates the full unitary group on the oscillators. A universal CV machine thus needs exactly

$$\underbrace{\text{degree 2}}_{\text{Gaussian / Clifford}} \quad + \quad \underbrace{\text{degree 3}}_{\text{cubic, non-Gaussian}} \quad .$$

The substrate supplies precisely these, as its two lowest matter invariants and no others below degree three: the *degree*-2 symplectic form on $\mathbb{F}_3^4$ that $\mathrm{Sp}(4,3)$ preserves — whose oscillator-rep limit is the Gaussian group $\mathrm{Sp}(4,\mathbb{R})$ that the photon's tritter, phase plate and modulator already realise (§6.5) — and the *degree*-3 $E_6$ Cartan cubic on the matter $27 = 3 \otimes 3 \otimes 3$, the $\varepsilon$-cubic preserved by $\mathfrak{sl}(3,\mathbb{F}_3)^3$ (`w33_paper.tex`, the 27-cubic and its 45 triads). So the paper's earlier statement that "the matter shell supplies the magic" is, in the CV picture, exact: the magic is the degree-3 $E_6$ cubic, the minimal non-Gaussian element, matching the Lloyd–Braunstein criterion on the nose. The substrate's invariant degrees $\{2,3\}$ *are* the universal CV generating set $\{\text{Gaussian}, \text{cubic}\}$. Together with the lattice tower $A_2 \subset D_4 \subset E_8$ this closes the architecture: $W(3,3)$ fixes both the fault-tolerant code and the universal gate set, leaving no free choice in the logical layer. (Witness: `analysis/w33_cv_universality_cubic.py`.)

Finally, the code and the gate set are not two independent facts: they are welded by a single group. The logical Clifford group of a GKP code on $n$ qudits of dimension $d$ is the symplectic group $\mathrm{Sp}(2n,\mathbb{Z}/d)$ over the logical phase space. For the holonet ($n = 2$, $d = 3$) this is

$$\mathrm{Sp}(4,\mathbb{Z}/3) = \mathrm{Sp}(4,3), \qquad |\mathrm{Sp}(4,3)| = 51840,$$

which is *exactly* $\mathrm{Aut}(W(3,3))$, the 2-qutrit Clifford group mod Pauli, and the Clifford group the photon's own tritter, phase plate and modulator already generate (§6.5, "symplectic closure = 51840"). So the substrate's symmetry group, the machine's realised gate group, and the GKP code's logical gate group are *one group*: the gates act on the code by construction. The code-preserving (transversal) Gaussian gates are the lattice automorphisms $\mathrm{Aut}(D_4) = W(F_4)$ of order $192 \cdot 6 = 1152$, whose $S_3$ factor is $D_4$ triality — the three-generation structure — and the remaining $[\mathrm{Sp}(4,3) : \mathrm{Aut}(D_4)] = 45$ logical Cliffords are teleported using the cubic resource. The logical layer of the holonet is thus a single coherent object, every part of it fixed by $W(3,3)$. (Witness: `analysis/w33_gkp_clifford_coherence.py`.)

### B.4 The code lattices are the moonshine ladder

The same lattices carry one last structure that ties the machine to the deepest arithmetic of the substrate. The holonet's modes are free bosons, and a GKP code is a free-boson theory compactified on its stabiliser lattice; an even lattice $\Lambda$ of rank $r$ is in turn the lattice vertex operator algebra (VOA) $V_\Lambda$ of central charge $c = r$ (Frenkel–Kac–Segal), while the stabiliser code itself fixes the corresponding (Narain) code CFT [23, 24]. The holonet's GKP code lattices are the substrate tower $A_2, D_4, E_8$ of ranks $2, 4, 8$, hence the lattice VOAs of central charge

$$c(A_2) = 2, \qquad c(D_4) = 4, \qquad c(E_8) = 8.$$

The top rung, the $E_8$ VOA at $c = 8$, is exactly the base of the moonshine ladder that $W(3,3)$ already realises on the physics side,

$$V_{E_8}\ (c = 8) \ \longrightarrow\ V_{\Lambda_{24}}\ (c = 24) \ \longrightarrow\ V^\natural\ (c = 24,\ \mathrm{Aut} = \mathbb{M}),$$

with $V^\natural$ the Frenkel–Lepowsky–Meurman Monster module (24 free bosons on the Leech torus, $\mathbb{Z}/2$-orbifolded), and $\Lambda_{24} \supset E_8^3$ the gauge lattice thrice. So the *same* $E_8$ is the holonet's optimal 4-mode

GKP code, the substrate's homology/gauge lattice, and the $c = 8$ chiral VOA that begins moonshine; and the top central charge $c = 24 = f = \chi(K3)$ is the $W(3,3)$ matter count. The computer's error-correcting code and the Monster module are one vertex-algebra tower — the architecture and the moonshine residual (R2) are the bottom and top of a single ladder of substrate lattices. (Witness: `analysis/w33_gkp_voa_bridge.py`.)

### B.5 What is physically needed: demonstrator versus fault-tolerant machine

It is worth being exact about what we are building, because the design is now complete but the *physical* realisation has two genuinely different layers. The single self-entangled photon of the build sheet — polarisation + time-bin qutrits, a tritter ($F_3$), a delay ladder and an EOM, read out by single-photon detectors — realises the *ideal, un-encoded* logical layer: the $\mathrm{Sp}(4,3)$ gate set, the routing network, and every parameter-free witness. It is buildable with today's quantum optics and it demonstrates the *logic*. But a single photon has fixed photon number; it cannot carry a GKP grid state (a many-photon quadrature lattice). So the photonic demonstrator is not, and cannot be, fault-tolerant.

Fault tolerance is the standard two-layer continuous-variable stack — and here both layers are substrate-fixed, with nothing left to choose:

$$\underbrace{\text{240 squeezed modes}}_{\text{physical}} \to \underbrace{\text{120 } D_4 \text{ GKP pairs}}_{\substack{\text{inner: analog}\to\text{digital} \\ \text{gain 1.5 dB}}} \to \text{240 GKP qutrits} \to \underbrace{[[240,81,4]]_3}_{\text{outer: Steinberg}} \to \text{81 logical qutrits.}$$

The inner code is the GKP code on the matter-shell lattice $D_4$ (converting continuous displacement noise into discrete qutrit errors, with the 1.5 dB coding gain that eases the squeezing budget); the outer code is the $[[240,81,4]]_3$ Steinberg code on the substrate's 240 edges. What remains is purely physical, and it is the *universal* CV fault-tolerance bottleneck, not a $W(3,3)$-specific one:

1. squeezed light at threshold ($\sim 10$ dB; protocol-dependent, 7–20 dB in the literature), lowered by the $D_4/E_8$ coding gain (1.5/3.0 dB);
2. GKP qutrit *state generation* — measurement-based breeding from squeezed/cat states with photon-number-resolving detection, or matter-assisted preparation: the hard step;
3. the cubic non-Gaussian resource — the degree-3 $E_6$ "matter-shell magic" — via a $\chi^{(3)}$ medium or magic-state injection;
4. a programmable beamsplitter/phase/squeeze network implementing the $\mathrm{Sp}(4,3)$ Clifford (mature integrated photonics);
5. homodyne detection for GKP and Steinberg syndrome readout.

This is the precise, honest statement of what the holonet is and what it still needs: a concatenated $\mathrm{GKP}(D_4) \circ \mathrm{Steinberg}\ [[240,81,4]]_3$ continuous-variable qutrit computer in which $W(3,3)$ fixes *both* code layers, the gate set ($\mathrm{Sp}(4,3)$ Gaussian + $E_6$ cubic) and the network — leaving *zero design freedom above the hardware.* The only open problem is the same one the entire CV fault-tolerance field faces, GKP-state generation at threshold squeezing, and the substrate's optimal lattices ease rather than remove it. (Witness: `analysis/w33_holonet_physical_stack.py`.)

## B.6 The gates are geometric: holonomic Clifford from the gauge connection

There is a third, gate-level layer of robustness that the substrate supplies for free, and it explains the name. The substrate gauge connection is flat along the 240 edges and curved on the matter graph, with binary-tetrahedral $2T$ holonomy. That holonomy group is not arbitrary:

$$2T \;\cong\; \mathrm{SL}(2,3) \;=\; \mathrm{Sp}(2,\mathbb{Z}/3), \qquad |2T| = 24,$$

the centre being $\{\pm I\}$ and the quotient $2T/\{\pm I\} = A_4$; and $\mathrm{Sp}(2,\mathbb{Z}/3)$ is precisely the single-qutrit Clifford group modulo Pauli. So the gauge connection's holonomy *is* the qutrit Clifford group, and the gates may be realised as *non-abelian holonomies* (Wilczek–Zee geometric phases) traced in the connection [4, 5]. Holonomic gates depend only on the geometric path, not on pulse timing or shape, so they are intrinsically robust to control error: this is fault tolerance at the *gate* level, complementary to the GKP code (analog level) and the Steinberg/color outer code (logical level). The full two-qutrit Clifford $\mathrm{Sp}(4,3)$ (order 51840) is generated by the matter-graph holonomy on each qutrit together with the entangling network.

The deeper point is the identity of the connection. The *same* connection whose *curvature* furnishes the substrate's gauge fields furnishes, through its *holonomy*, the computer's gates: the gauge interactions of the physics and the logic gates of the machine are one and the same non-abelian holonomy. This is the literal content of "holo-net". (Witness: `analysis/w33_holonomic_gates.py`, constructing $\mathrm{SL}(2,3)$ and verifying order 24, symplecticity, centre $\{\pm I\}$, quotient $A_4$, and the order spectrum $\{1,1,8,6,8\}$ of $2T$.)

## B.7 A near-term falsification test: the demonstrator as an experiment on the substrate

The single-photon demonstrator of the build sheet is buildable now and needs no GKP states or squeezing, yet the holonomic-gate and quasicrystal-clock structure make it a *discriminating, parameter-free* test of the substrate hypothesis itself. Three signatures, all measurable on that machine:

1. **Geometric-gate timing-independence (the discriminator).** If the gates are Wilczek–Zee holonomies (the substrate claim), the gate unitary depends only on the *path* traced in control space, not on the schedule. Reparametrising the loop — any speed, any pulse shape with the same path — leaves the gate, read out through the trace–Choi visibility $V(U) = |\mathrm{Tr}\, U|/3$, *invariant*; a generic dynamical gate would change. A measured timing-dependence falsifies the holonomic claim.

2. **Holonomy-group fingerprint.** Composing the geometric gates must close into $2T = \mathrm{SL}(2,3)$: order 24, element-order spectrum $\{1,1,8,6,8\}$, and a single-qutrit Clifford visibility spectrum $V \in \{0, \frac{1}{3}, \frac{1}{\sqrt{3}}, 1\}$ (with $V(F_3) = \frac{1}{3}$ and $V(X) = V(Z) = 0$). Gate-set tomography reads this out; any other group falsifies the gauge-connection-holonomy identification.

3. **Provably aperiodic clock.** The Boerdijk–Coxeter drive rotates by $\theta = \arccos(-\frac{2}{3}) \approx 131.81^\circ$ per pass. By Niven's theorem [15] the only rational $\cos(r\pi)$ with $r$ rational are $\{0, \pm\frac{1}{2}, \pm 1\}$, and $-\frac{2}{3}$ is none of them, so $\theta/\pi$ is irrational and the drive *never recurs* — a provably aperiodic time-quasicrystal clock. A measured recurrence at any finite pass falsifies it.

Each signature is fixed by the geometry with no free parameter, and each is within reach of the demonstrator's existing optics. Together they turn the buildable machine into an experiment *on the substrate*: not a test of whether the device computes, but of whether the gauge connection, the holonomy group, and the quasicrystal drive are the ones $W(3,3)$ predicts. (Witness: `analysis/w33_near_term_falsification_test.py`.)

### B.8 BT1402 runtime-frontier handoff: readout, port, and certificate

The post-BT1377 runtime frontier makes the physical story sharper. The deterministic packet machine is a protected Clifford machine first: the 8-tick optical word, Q6/tomotope packet edge, $S_3$ phase synchronization, central $C_3$ Steinberg scheduler, and 51840-window $\mathrm{Sp}(4,3)$ supercycle are all finite Clifford data. Universal computation enters at the non-Clifford boundary, not by pretending that this Clifford kernel is already universal by itself.

The near-term single-photon demonstrator now has a concrete readout contract. Its Bell-qutrit signatures are

$$V(I) = 1, \qquad V(F_3) = \frac{1}{3}, \qquad V(X) = V(Z) = 0,$$

and route-controlled Clifford transport deliberately hides route coherence if the Bell legs are discarded: the reduced route register has uniform probabilities, purity 1/3, and zero $\ell_1$ coherence. Coherence is restored only by the quantum eraser on the Bell branch,

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}},$$

which succeeds with probability 1/3, restores route $\ell_1$ coherence 2, and routes the interferometer to a single output port. The current parametric noise envelope keeps this falsifiable: the baseline readout has $\gamma = 0.92887$ and port-0 probability 0.95258, while the conservative case keeps $\gamma = 0.79198$ and port-0 probability 0.86132.

The non-Clifford resource has likewise become an ABI rather than a slogan. The Hesse-SIC/T token has nine outcomes, is consumed at a Clifford packet boundary, and feeds a Clifford correction plus one $T$-frame parity bit into the next 8-tick word. Over one 51840-tick Clifford window there are 720 microframes. In the current queue envelope the baseline heavy case still has positive slack (about 55.75 successful tokens per window), and the conservative medium case still has positive slack (about 89.51 tokens per window). This is an engineering envelope, not a claim that the SIC optics or magic-state yield are solved.

Finally, the correction frontier is now stated in certificate language. The best current $S_3$ gauge witness has score 210 and is locally strict through the checked radius, but the MaxSAT frontier remains witness-only: the example optimality certificate exercises the verifier path and is explicitly synthetic. A project-level global optimum requires an imported solver upper-bound proof matching the witness score. This is the honest architecture boundary after BT1401: the single-photon experiment tests the Bell/route physics, the Clifford machine runs the protected finite kernel, the Hesse-SIC/T port supplies the non-Clifford ABI, and the remaining global optimality claim is a certificate import problem.

Witnesses: `tools/bt1394_reduced_qutrit_demonstrator.py`; `tools/bt1396_qutrit_quantum_erasure_readout.py`; `tools/bt1385_hesse_sic_t_port_abi.py`; `tools/bt1391_hesse_sic_t_queue_model.py`; `tools/bt1395_s3_maxsat_bound_pathway.py`; `tools/bt1400_qutrit_erasure_noise_sensitivity.py`.

### B.9 BT1403 eraser-lift port: the Hesse boundary is the readout boundary

The Hesse port is not a second computer attached to the holonet after the fact. It is the quantum-eraser measurement boundary lifted by one qutrit phase label. BT1396 gives the first half of the boundary: the route register is incoherent if the Bell legs are traced out, and coherence returns only when the Bell branch is erased by the projector

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}}.$$

This is a three-branch measurement, with branch labels

$$\Omega, \quad Z\Omega, \quad X\Omega.$$

BT1385 gives the non-Clifford half: the Hesse-SIC/T token has nine outcomes. The ABI statement is therefore the exact lift

$$9 = 3 \times 3, \qquad h = 3r + p,$$

where $r \in \{0, 1, 2\}$ selects the route/Bell branch and $p \in \{0, 1, 2\}$ is the added phase label. Equivalently, the nine Hesse outcomes are exactly 3 route branches times 3 phase labels. The feed-forward record maps the outcome to a two-trit Clifford correction $X^r Z^p$ plus the one $T$-frame parity bit needed by the non-Clifford injection.

This is the physical meaning of the non-Clifford port in the single-photon architecture. The same interferometric readout that proves the route qutrit is coherent also supplies the boundary at which a non-stabilizer qutrit resource can be consumed. The Clifford runtime remains an 8-tick packet machine; the Hesse measurement writes a side record, restores the packet ABI, and hands the next word back to the protected scheduler. Thus the architecture is not "Clifford machine plus mysterious magic box" but

$$\text{Bell-branch eraser} \quad \longrightarrow \quad \text{Hesse } 3 \times 3 \text{ lift} \quad \longrightarrow \quad \text{next Clifford packet word.}$$

The boundary remains honest: this identifies the ABI and alphabet of the port, not the optical SIC implementation or magic-state factory yield. (Witness: `tools/bt1403_hesse_port_eraser_lift.py`.)

## B.10 BT1404 holonet scope: one Hesse grid is one microframe

BT1404 makes the eraser-lift port visible as a packet oscilloscope. The timing identity is

$$9 \text{ Hesse outcomes} \times 8 \text{ packet ticks} = 72 \text{ ticks},$$

exactly one BT1385/BT1391 microframe. In plain terms: 9 Hesse outcomes times 8 packet ticks fill the whole scope. Thus the entire Hesse outcome alphabet fits in one microframe, with each cell $h = 3r + p$ occupying one 8-tick return word.

The scope word is:

| | |
|---|---|
| 0 | erase Bell branch |
| 1 | record route trit $r$ |
| 2 | record phase trit $p$ |
| 3 | apply $X^r$ frame correction |
| 4 | apply $Z^p$ frame correction |
| 5 | store $T$-frame bit $(r + p \bmod 2)$ |
| 6 | restore Clifford ABI |
| 7 | hand off next packet word. |

This is the first machine-surface form of the non-Clifford boundary: not just a nine-outcome label, but a one-microframe live trace compatible with the 720 microframes in a 51840-tick Clifford window. The generated static scope is `docs/bt1404_holonet_scope.html`. It remains an ABI display, not a claim of physical SIC-optics hardware.

## B.11 BT1405 continuous Q6 path router

BT1374 turned packet digits into executable Q6 edge addresses, but deliberately left one boundary open: an address list is not yet a continuous walk. BT1405 closes that boundary at the tomotope ABI level. For every compiled BT828 packet program, orient the packet edges to minimize the total Hamming connector length, insert shortest Q6 connector walks between consecutive packet edges, and map every traversed edge back through the BT1371 table.

The six-digit stress route keeps the six BT1374 packet edges

$$175,\ 133,\ 56,\ 37,\ 142,\ 77$$

but now they are waypoints on a single hypercube walk:

$$6 \text{ packet edges plus } 10 \text{ connector edges} = 16 \text{ Q6 steps.}$$

In words: 6 packet edges plus 10 connector edges give the continuous carrier trace. Since the tomotope body budget is 48 ticks, the continuous trace leaves

$$48 - 16 = 32$$

slack ticks inside the same body. Thus the route is no longer merely a set of six distinct Q6 addresses; it is a contiguous Q6 path from `110111` to `010011`, with every traversed edge carrying a tomotope flag and block. This is still an ABI route certificate, not a waveguide layout or detector-loss model. (Witness: `tools/bt1405_continuous_q6_path_router.py`.)

## B.12 BT1406 tomotope-body edge pulse scheduler

BT1406 uses the slack exposed by BT1405 as timing structure. Each Q6 edge in the continuous route is not a bare edge label; it expands into the ternary microcycle

| | |
|---|---|
| 0 | LOAD_FLAG: load tomotope flag, block, and transversal |
| 1 | FLIP_Q6_AXIS: perform the one-bit Q6 traversal |
| 2 | LATCH_VERTEX: latch the target vertex and ABI record. |

Therefore the stress route obeys

$$16 \text{ Q6 edge traversals} \times 3 \text{ pulse phases} = 48 \text{ body ticks.}$$

In words: 16 Q6 edge traversals times 3 pulse phases equals 48 body ticks. The old 32-tick slack becomes the timing shell: the stress route has no idle body tick after expansion. Its packet-edge load ticks are

$$0,\ 6,\ 12,\ 21,\ 27,\ 45,$$

so the final packet edge occupies ticks $45, 46, 47$: load tomotope flag 180 in block 45, flip $010111 \to 010011$, and latch the target vertex. This is a timing ABI, not a calibrated optical pulse-width, detector-jitter, or loss model. (Witness: `tools/bt1406_tomotope_body_edge_pulse_scheduler.py`.)

### B.13 BT1407 microframe transaction composer

BT1407 closes the whole oscillator frame. BT1406 fills the first 48 tomotope-body ticks. The remaining BT1300 local-lift epilogue is

$$24 = 3 \times 8,$$

exactly three Hesse return words. In words: 48 Q6 body pulse ticks plus 3 Hesse return words equals 72 ticks.

For the six-digit stress route, the final target digit is 4. Hence $4 \bmod 3 = 1$, so the selected Hesse route branch is $r = 1$. The epilogue is the phase fanout along that branch:

| frame ticks | $h$ | $(r,p)$ | correction |
|---|---|---|---|
| 48–55 | 3 | $(1,0)$ | $X^1Z^0$ |
| 56–63 | 4 | $(1,1)$ | $X^1Z^1$ |
| 64–71 | 5 | $(1,2)$ | $X^1Z^2$. |

Thus ticks 0–47 execute the Q6/tomotope carrier body, while ticks 48–71 execute the Hesse return words that erase the branch, record route and phase, apply the Pauli-frame correction, store the $T$-frame bit, restore the Clifford ABI, and hand off the next packet word. This is a full-frame ABI transaction, not a SIC-optics implementation. The witness is the BT1407 generator and JSON certificate.

### B.14 BT1408 Witting contextual communication bridge

BT1408 imports the Witting-polytope communication scheme into the same runtime ABI. The external protocol uses 40 Witting rays and 40 orthogonal tetrads. Around any selected ray the other side of the delayed-query round splits as

$$1 \text{ same ray} + 12 \text{ compatible orthogonal rays} + 27 \text{ incompatible rays} = 40.$$

In words: 1 same ray plus 12 compatible orthogonal rays plus 27 incompatible rays make the whole Witting communication desk. Hence the accepted key-agreement rate is

$$\frac{1+12}{40} = \frac{13}{40},$$

and the rejected sector is 27/40, the same $q^q$ matter shell that the holonet already uses as contextual fuel.

The four slots of an accepted Witting tetrad are not a new bus. They are the four local residues in the packet ABI:

$$\text{tomotope\_flag} = 4\,\text{tomotope\_block} + (\text{mirror\_slot} \bmod 4).$$

Thus an accepted communication round has the handoff

$$\text{Witting query pair} \to \text{common tetrad} \to \text{mirror slot} \to \text{Q6 body} \to \text{Hesse epilogue}.$$

The contextuality audit remains the corrected BT823 ceiling 36/40, with deficit 4 and contextual fraction 1/10; the older 34/40 value from the communication paper is a useful historical warning, not the local theorem. BT1408 is an ABI bridge, not a security proof or a ququart hardware certificate.

### B.15 BT1409 Witting duplex admission scheduler

BT1409 separates the two acceptance clocks inside the Witting communication bridge. The state-query clock is the BT1408 shell:

$$\frac{13}{40} = \frac{1 \text{ same ray} + 12 \text{ compatible rays}}{40}.$$

The basis-query clock is smaller. A selected Witting ray lives in exactly four tetrads, hence

$$\frac{4}{40} = \frac{1}{10}$$

is the witness aperture. In words: 13/40 is communication throughput, while 1/10 is contextual witness aperture. The complementary basis shadow is

$$\frac{36}{40},$$

which matches the corrected BT823 noncontextual ceiling. Thus the old "36/40" number is not a communication success rate; it is the basis retry shadow and the classical boundary.

For one selected ray, serving every compatible state as a BT1407 transaction uses

$$13 \times 72 = 936$$

frame ticks. Auditing only the four witness apertures uses

$$4 \times 72 = 288$$

frame ticks. Rejected choices need not consume a full frame; they are classified as retry shadow or matter-shell context unless an implementation chooses to log them. This is a scheduler certificate, not a cryptographic security proof.

### B.16 BT1410 Witting delayed-query frame compiler

BT1410 makes the delayed-query step operational. The protocol in [25] first lets the two sides query Witting rays, then delays the final basis measurement until the queried states have been compared. The holonet reading is a compiler:

$$(\text{Alice ray}, \text{Bob ray}) \longmapsto \text{common Witting tetrad} \longmapsto \text{BT1407 frame}.$$

If the two rays share no tetrad, the pair is a retry-shadow event and no full frame is opened. If they are distinct compatible rays, they share exactly one tetrad, so the selected basis is forced. If they are the same ray, they share four tetrads; a two-bit aperture selector chooses which witness basis to audit.

There are two tables, and confusing them loses the physics. The logical ordered pair table has

$$40 \cdot 40 = 1600$$

rows, split as

$$40 \text{ same-ray pairs} + 480 \text{ compatible distinct pairs} + 1080 \text{ retry-shadow pairs}.$$

Thus the accepted logical table has 520 rows and rate $520/1600 = 13/40$. The physical frame table is basis-local:

$$40 \text{ tetrads} \times 4 \text{ Alice query slots} \times 4 \text{ Bob query slots} = 640.$$

Equivalently, every Witting tetrad is a $4 \times 4$ frame tile with four diagonal witness apertures and twelve off-diagonal data handshakes. Across all forty tetrads this is

$$160 \text{ diagonal witness records} + 480 \text{ off-diagonal data records}.$$

The extra $640 - 520 = 120$ basis-local records are not extra throughput. They are the same-ray contextual aperture: each of the 40 rays has four basis contexts, so the physical compiler carries three extra context choices per ray above the single logical same-ray identity.

Once the common basis is selected, the later four-outcome measurement slot is just the packet slot:

$$\texttt{tomotope\_flag} = 4\,\texttt{tomotope\_block} + (\texttt{mirror\_slot} \bmod 4).$$

Therefore the Witting desk is now a packet admission ROM. Off-diagonal entries open communication frames; diagonal entries open contextual audit frames; both land in the same BT1407 72-tick body-plus-Hesse-epilogue transaction. This matches the feed-forward/programmable-loop physical idiom of time-bin photonic qudit processors [26], but BT1410 itself remains a finite compiler certificate rather than a loss model, detector calibration, or cryptographic security proof.

### B.17 BT1411 Witting basis analyzer unitaries

BT1411 supplies the physical analyzer that BT1410 deliberately left abstract. For every Witting tetrad

$$B = (r_0, r_1, r_2, r_3)$$

define $U_B$ by taking row $j$ to be the conjugate ray $r_j^\dagger$. Then

$$U_B\, r_j = e_j,$$

so detector slot $j$ is exactly the packet residue $\texttt{mirror\_slot} \bmod 4$. This is the standard linear-optics measurement move: an analyzer unitary rotates the chosen orthonormal basis to the detector basis, after which fixed detectors read the slots. General finite-dimensional unitaries are optically synthesizeable by beam-splitter/phase-shifter meshes [27, 28]; BT1411 shows the Witting case is much more structured than a generic $4 \times 4$ mesh.

The forty analyzers have the exact hardware split

$$1 + 12 + 27.$$

There is one computational direct-rail analyzer, twelve one-direct-rail plus complement-tritter analyzers, and twenty-seven four-three-rail contextual analyzers. Equivalently, the nonzero-entry histogram is

$$4^1, \quad 10^{12}, \quad 12^{27},$$

where the exponent records how many analyzers have that many nonzero matrix entries. No Witting analyzer uses all 16 entries of a generic dense four-mode unitary. The entry alphabet is only

$$0, \quad 1, \quad \frac{\pm 1}{\sqrt{3}}, \quad \frac{\pm\omega}{\sqrt{3}}, \quad \frac{\pm\omega^2}{\sqrt{3}}, \qquad \omega^3 = 1.$$

Thus the physical reading is sharper than "program a universal interferometer." The Witting communication desk is a sparse analyzer ROM:

$$\begin{aligned} \text{delayed-query tetrad} &\longmapsto \text{four-mode Witting analyzer} \\ &\longmapsto \text{detector slot} \longmapsto \text{BT1374 packet residue.} \end{aligned}$$

The twelve middle analyzers literally have one bypass rail and a three-rail tritter-like complement; the twenty-seven contextual analyzers are balanced three-rail rows, each dark on one rail. This is a physics certificate for the single-photon ququart analyzer bank, not a chip calibration, loss budget, or beam-splitter-angle synthesis.

### B.18 BT1412 toroidal Q4 oscillator boundary

BT1412 returns to the $4 \times 4$ toroidal square because that square is not a decorative drawing of the local packet router. With ordinary boundary conditions the knight graph has only 24 edges and mixed degree; with toroidal wraparound it is exactly $Q_4$. The existing closed toroidal knight tour is therefore a 16-tick Gray Hamilton clock on the four-cube: each packet tick flips one $Q_4$ bit, and the board parity $(r + c)$ mod 2 is the same as the $Q_4$ word parity. Hence parity alternates on every clock edge.

The new splice is that the two-dimensional cells of this same local router have the right oscillator boundary:

$$\#F_2(Q_4) = \binom{4}{2} 2^2 = 24 \text{ Q4 square faces} = q(q-1)(q+1).$$

BT802 already proved that the tetrahedral oscillator is allowed exactly at the mod-12 marks

$$\{0, 3, 4, 7\}$$

and that the neighborly genus level $h = q = 3$ is forbidden by the discriminant 145. Lifting those four marks onto the 24-plaquette $Q_4$ shell gives the exact aperture

$$\frac{4}{24} = \frac{1}{6},$$

which is the local "one-sixth" selector window in the clock face, not a fitted probability. Removing the forbidden genus level then lands on the toroidal dual edge invariant:

$$24 - 3 = 21.$$

That 21 is not numerology. It is the edge channel preserved by the dual toroidal polyhedra:

$$\text{Császár} : (V, E, F) = (7, 21, 14), \qquad \text{Szilassi} : (V, E, F) = (14, 21, 7).$$

Császár is the maximal vertex-adjacency boundary, Szilassi is the maximal face-adjacency boundary, and geometric duality swaps $V$ and $F$ while keeping $E = 21$ intact. In this exact local reading the tetrahedral oscillator sits between the two toroidal boundaries: the $Q_4$ plaquette shell supplies the 24-slot face clock, the forbidden $q = 3$ genus gap removes the non-realizable handle, and the surviving 21-channel edge bus is precisely the dual boundary shared by the two toroidal polyhedra.

The error-code boundary is also now sharper. A Hamilton cycle in a labeled hypercube is a cyclic Gray code, but a snake-in-the-box code is an induced path or cycle constraint [11, 12]. The full BT1412 $Q_4$ Gray clock is not claimed to be such a snake: it uses 16 cycle edges while $Q_4$ has

32 edges, so there are 16 non-cycle chords. The protected projection is instead the every-other-tick even layer

$$0,\ 6,\ 3,\ 5,\ 9,\ 15,\ 10,\ 12,$$

an eight-word parity-even code whose cyclic steps have Hamming distance 2 and whose pairwise minimum distance is 2. Thus the physical story is: toroidal $Q_4$ Gray clock for packet motion, every-other parity projection for single-bit-error detection, and the 1/6-aperture oscillator boundary tying the 24-plaquette shell to the 21-edge Császár/Szilassi channel. BT1412 is an ABI certificate, not a waveguide layout, a calibrated oscillator, or a new snake-code optimum. (Witness: `tools/bt1412_toroidal_q4_oscillator_boundary.py`.)

### B.19 BT1413 Q4 plaquette-tomotope compiler

BT1413 makes the 24-plaquette boundary executable. BT1363 had already shown that Q4 face-edge incidence modulo the antipodal translation is the tomotope/Reye medial layer, with 48 middle blocks. BT1371 had already shown that the 192 tomotope flags are a bijective Q6 edge address table. The BT1413 compiler is just the composed map:

$$24 \text{ Q4 plaquettes} \cdot 4 \text{ edge incidences}/2 = 48 \text{ middle blocks},$$

$$48 \text{ middle blocks} \cdot 4 \text{ flag residues} = 192 \text{ tomotope/Q6 flags}.$$

The generated certificate checks that each middle block has exactly two Q4 lifts, that the three ternary sheets carry 64 flags each, that each sheet hits all 16 tomotope face labels exactly once at block level, and that every flag is a one-bit Q6 edge address. The result is not an optical layout. It is the local address compiler that lets the toroidal Q4 shell talk directly to the tomotope/Q6 packet ABI.

### B.20 BT1414 Csaszar-Szilassi dual port

BT1414 uses the active/ground split that BT1317 already found in the tomotope packet:

$$192 = 168 + 24.$$

The active part is now a concrete dual analyzer port:

$$21 \text{ shared toroidal edge channels} \cdot 2 \text{ orientations} \cdot 4 \text{ residues} = 168.$$

In Császár mode the seven analyzers are vertex-neighborhood checks: each vertex of $K_7$ sees six incident edge channels, and each channel is seen by two vertex analyzers. In Szilassi mode the same 21 channels are read as the dual complete face-adjacency checks: seven face analyzers, again six channels each, again two analyzers per channel. The current BT1318 metric-axis records fix Császár vertex 6 and Szilassi face 4, so the crossed calibration channel is the shared edge $\{4, 6\}$.

The remaining 24 flags are not discarded. They are the BT1412/BT1413 Q4 plaquette guard band. Thus the local physical port has a sharp shape:

$$21 \cdot 2 \cdot 4 \quad \text{active dual-boundary slots} \quad + \quad 24 \quad \text{Q4 plaquette guards}.$$

### B.21 BT1415 even-projection Steinberg/CSS syndrome ledger

BT1415 finishes the front end by making the every-other Q4 projection into a check ledger. The allowed clock states are the eight even words

$$0,\ 6,\ 3,\ 5,\ 9,\ 15,\ 10,\ 12.$$

They form a binary distance-2 parity layer: every allowed word has parity zero, and any one-bit Q4 clock fault flips the syndrome bit. This check is binary front-end logic; the memory register is still the existing $\mathbb{F}_3$ Steinberg/CSS object.

BT1375 supplies the memory clock. The central $C_3$ action on the Steinberg register has 27 three-cycles and nilpotent rank profile

$$54,\ 27,\ 0.$$

Therefore the exact syndrome ledger is

$$27 \text{ Steinberg central cycles} \cdot 8 \text{ even Q4 states} = 216,$$

and the BT1414 guard band completes the existing W33 CSS edge ledger:

$$216 + 24 = 240.$$

This is the first complete local front end in the paper:

$$\begin{aligned} \text{Q4 plaquettes} &\to \text{tomotope/Q6 flags} \\ &\to \text{Császár/Szilassi port} \\ &\to \text{even-parity syndrome} \\ &\to \text{Steinberg/CSS edge ledger.} \end{aligned}$$

I also checked the external Golden Quartic/Moebius-ball electron line of work. Otto's 2022 paper connects a golden quartic, icosahedral coefficients, chirality, and a proposed Moebius-ball electron geometry [33]; his 2025 preprint asks whether AI can compare such extended electron models [34]. A 2025 review frames this family as Zitterbewegung/field-dynamics modelling rather than established particle physics [35]. The holonet use is deliberately conservative: closed-loop topology and chirality are good design warnings, but the repo's exact quartic frontier remains the two independent $D_4$ quartic atoms in the local minimal-magic audit. No electron mass, spin, or waveguide claim depends on the Moebius-ball model. (Witnesses: `tools/bt1413_q4_plaquette_tomotope_face_compiler.py`, `tools/bt1414_csaszar_szilassi_dual_physical_port.py`, and `tools/bt1415_even_projection_steinberg_syndrome_layer.py`.)

### B.22 BT1416 sparse CSS intertwiner

BT1416 turns the BT1415 ledger from a count into matrices. It rebuilds the canonical W33 edge-chain CSS carrier:

$$H_X = d_1 :\ C_1 \to C_0, \qquad H_Z = d_2^T :\ C_1 \to C_2,$$

over $\mathbb{F}_3$. The generated sparse certificate has shapes

$$H_X \in \mathbb{F}_3^{40\times 240}, \qquad H_Z \in \mathbb{F}_3^{160\times 240},$$

and verifies

$$\operatorname{rank}(H_X) = 39, \qquad \operatorname{rank}(H_Z) = 120, \qquad H_X H_Z^T = 0.$$

Thus the carrier is still the exact edge code

$$[[240,\ 240-39-120,\ 3]]_3 = [[240, 81, 3]]_3.$$

The new map is the typed $240 \times 240$ ledger interwiner:

$$\Pi(e_i) = e_i, \qquad 0 \le i < 240.$$

Rows $0, \ldots, 215$ are the $\mathbb{F}_2$ Q4 parity front end, with check vector $(1, 1, 1, 1)$, rank 1, and the eight even Q4 words as a distance-2 kernel. Rows $216, \ldots, 239$ are the guard tail addressing the same $\mathbb{F}_3$ edge carrier. This is deliberately not a field coercion: the binary clock syndrome and ternary CSS stabilizer live in different fields, and the interwiner is the finite scheduling boundary between them. (Witness: `tools/bt1416_css_sparse_intertwiner_matrices.py`.)

### B.23 BT1417 linear-optical dual-port primitive synthesis

BT1417 converts the BT1414 port into an objectwise single-photon optical primitive list. The active dual port is not a vague optical metaphor; it has the exact inventory

21 edge-channel couplers, 42 oriented phase latches,
168 active detector bins, 24 guard apertures.

The 21 couplers are the shared $K_7$ edge channels. The 42 latches choose the two directions on each channel. The four residue bins behind each oriented latch are the tomotope flag residue. The 24 Q4 guard apertures are separate from the active mesh.

The analyzer matrices are $7 \times 21$ incidence matrices. In Császár mode a row is a vertex star; in Szilassi mode a row is the dual face star. In both modes the Gram law is

$$AA^T = 5I_7 + J_7,$$

so every analyzer sees six channels and any two analyzers overlap in exactly one channel. This is the chip-level object language the previous ABI was missing: coupler, direction latch, residue demultiplexer, detector bin, and guard aperture. It is still not a calibrated waveguide mask, loss budget, or detector-efficiency model. (Witness: `tools/bt1417_linear_optical_dual_port_primitives.py`.)

### B.24 BT1418 finite D4-quartic magic injection

BT1418 closes the non-Clifford aperture without importing the continuum Moebius-ball electron hypothesis. The repo-local minimal-magic audit already found the exact signed frontier: two independent irreducible $D_4$ quartic atoms, with no shared quadratic subfield and splitting field

$$D_4 \times D_4.$$

The BT1415/BT1416 guard band fits those atoms exactly:

$$2 \text{ quartic atoms} \cdot 4 \text{ algebraic branches} \cdot 3 \text{ qutrit phases} = 24 \text{ guard apertures.}$$

Then the internal $D_4$ orientation torsor expands each aperture to the tomotope flag bus:

$$24 \text{ guard apertures} \cdot 8\ D_4 \text{ orientations} = 192 \text{ oriented tomotope tokens.}$$

This is the exact finite counterpart of the golden-quartic topology that is safe to use in the paper. For one quartic atom, the Steinberg lift gives

$$4 \text{ branches} \cdot 27 \text{ central cycles} \cdot 8\ D_4 \text{ orientations} = 864,$$

which is exactly the repo's Golden $D_4$/Weyl shell. With both independent atoms the shell is 1728, but the two atoms stay independent; BT1418 does not collapse them into one continuum object. The physical reading is that the 24 guard apertures are the non-Clifford injection rail, while the 216 parity rows remain the cheap binary front-end syndrome and the $\mathbb{F}_3$ CSS carrier keeps its commutation proof. (Witness: `tools/bt1418_d4_quartic_magic_injection_frontier.py`.)

### B.25 Self-error-correction and why the primitive is one

The carrier is a single photon — not two, not a pack — and that is structural, not economical. Two ingredients explain why the universe's primitive is one.

*Self-entanglement instead of inter-particle entanglement.* A photon carries several internal registers (polarization, path, time-bin, frequency, transverse mode), and entanglement among them needs no second particle. Two ternary internal registers already give $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$ — the entire two-qutrit substrate register *inside one photon*. The geometry is realised within a single indivisible quantum, which is exactly the operator–operand self-duality of §2.

*Self-error-correction, and no-cloning as a feature.* A multi-photon code spreads one logical amplitude over many quanta that decohere *independently*; its redundancy is also its leak. A single photon cannot be cloned — and here that is the asset, not the obstacle: the logical amplitude is one indivisible excitation, with no independent copies to dephase against each other. Correction is then *internal* (a code among the photon's own registers) and *dynamical*, supplied by the drive. The Boerdijk–Coxeter loop (§11) is not a periodic clock but a genuine *time quasicrystal*: its twist $\theta = \arccos(-\frac{2}{3})$ is substrate-fixed — at $q = 3$, $-\frac{2}{3} = -(q-1)/q = 2 \cdot (-1/q)$, the pairwise vertex dot of the regular $q$-simplex (the tetrahedron = the $q$=3 simplex) — and the stroboscopic orbit $\{n\theta \bmod 2\pi\}$ obeys the Steinhaus three-gap law at every $N$ (exactly two gaps at $n = 30 = h(E_8)$), is Weyl-equidistributed, and never recurs (Niven). It therefore carries two incommensurate frequencies $(2\pi,\ \theta)$, a pure-point spectrum: the quasiperiodically-driven (Fibonacci-type) setting whose emergent dynamical symmetry topologically protects an edge mode [6, 8]. The photon self-protects its logical qutrit through its own substrate-fixed drive — no copies, no external apparatus. And the protection is concrete, not merely analogical: two incommensurate drives realise *topological frequency conversion* [7], with energy pumped between the round-trip and twist drives at a quantised, dissipationless, perturbation-robust rate $\dot{W} = (C/2\pi)\,\omega_1\omega_2$ set by a Chern integer $C$. We compute it. Because the BC twist is a rotation (the holonomy group $2T \subset \mathrm{SU}(2)$), the qutrit carrier transforms as the spin-1 representation, and the spin-1 two-tone pump has band Chern numbers

$$(C_-, C_0, C_+) = (+2,\ 0,\ -2)$$

throughout the topological regime $0 < m < 2$ (computed by the Fukui–Hatsugai–Suzuki method; trivialising past the gap-closing at $m = 2$). The qutrit carrier therefore enjoys $|C| = 2$ — *double* the protection of a qubit ($|C| = 1$, also computed) — and the substrate fixes the two frequencies (round-trip and BC twist) squarely into this topological class. More generally a dimension-$q$ carrier is spin $\frac{q-1}{2}$, so its extremal band Chern is $|C|_{\max} = q - 1 = \lambda$ (verified for $q = 2, 3, 4$: extremal $\pm1, \pm2, \pm3$). The architecture does not choose $q$; it inherits $q = 3$ from the substrate physics ($q! = 2q$, KO-dimension $2q = 6$) and reaps the enhanced protection as a consequence. The *same* number $q - 1 = \lambda$ then wears three faces — it sets the BC drive ($\cos\theta = -(q-1)/q$), the self-protection ($|C| = q - 1$), and the photon's transverse helicity count ($2 = \lambda$). *(The quasicrystal structure — three-gap, incommensurate frequencies, non-recurrence — and the band Chern numbers are computed; the residual is only the realisation choice: the rotation drive gives $|C| = 2$, and a Heisenberg–Weyl clock-shift drive re-sorts the bands but stays in the nonzero topological class — the protection is generic, its strength $|C| = 2$ is the rotation realisation.)* (Witnesses: `analysis/w33_self_correction_quasicrystal.py`, `analysis/w33_topological_pump_test.py`, `analysis/w33_qutrit_topological_pump.py`.)

### B.26 Why the primitive is one *massless* photon

One more layer answers *why a photon* and not some other quantum. Wheeler's one-electron universe (relayed to Feynman in 1940) imagined every electron as a single worldline weaving forward and backward through time. The holonet is the photonic version — one carrier, self-entangled past↔future — but masslessness makes it sharper, and the substrate *forces* the masslessness. Three facts lock together.

*Null worldline, zero proper time.* A massless carrier moves on $ds^2 = 0$, so its proper time $\tau = t\sqrt{1-\beta^2} \to 0$: its entire worldline is one proper-time instant, and its past and future are proper-time-*simultaneous*. That is exactly what lets a single photon entangle its past and future registers — a relation at one proper-time point. Wheeler's massive electron must literally weave; the massless photon is all-at-once. Self-reference wants masslessness.

*Gauge invariance forces it.* The photon is the gauge boson of the substrate connection of §6 — the one whose holonomy is the Clifford gates ($2T = \mathrm{SL}(2,3)$) and whose curvature is the gauge fields. The substrate gauge symmetry is unbroken (the holonomy is the full Clifford group), and gauge invariance forces an unbroken gauge boson to be massless, removing the longitudinal mode (Wigner's classification).

*The helicity count is the substrate's.* A massless vector has Wigner little group ISO(2) and exactly *two* transverse helicities $\pm 1$; a massive vector has SO(3) and *three*. The carrier therefore has $2 = \lambda = q - 1$ polarisations — precisely the paper's photon-polarisation count $\lambda$ — and only a massless carrier gives it. (Its clock is then the optical frequency: $\tau = 0$ but the phase $\phi = \omega t$ still accrues, so the time-bin/frequency registers — the very degrees of freedom that self-entangle — are the photon's internal clock. A testable corollary: in vacuum the self-entanglement carries no proper-time decoherence, i.e. its visibility is worldline-length independent up to lab-frame loss and dispersion.)

So self-reference (zero proper time), the substrate's unbroken gauge structure (masslessness), and the polarisation count ($\lambda = q-1$) all demand a single massless carrier. The universe's primitive is one photon. (Witness: `analysis/w33_why_one_massless_photon.py`.)

### B.27 The single carrier, the geon, and the fractal of nested shells

Three pictures of the same machine fall out, each tested. *(i) Minimal activation.* If one carrier's worldline braids through the substrate's topology to compute, how few controls suffice? The gate group is $\mathrm{Sp}(4,3)$ (order 51840), and it is 2-generated: a search over the natural symplectic "switches" (one-mode Fourier/phase, two-mode $CZ/SUM$) finds 13 generating *pairs* — e.g. $\langle F_2,\ F_1 \cdot CZ\rangle$ reaches all 51840. So flipping *two well-chosen switches* in the right temporal pattern (a binary word, scheduled by the Boerdijk–Coxeter quasicrystal) reaches every gate; the naive pair $\langle F_1, CZ\rangle$ does *not* (it generates only a 108-element subgroup), so the pair must be a generating one. Moreover the patterns are *short*: the Cayley diameter of $\mathrm{Sp}(4,3)$ under such a pair is only 14, so every one of the 51840 gates is a word of at most 14 switch-flips (mean $\approx 10$), and the diameter is the same across the generating pairs tested — the worst-case compilation length is substrate-fixed, with the hardware entering through *which* operations are the two switches, not through the length. The switches are two elementary braids of the carrier's worldline, robust because braiding depends only on the topology — the holonomic robustness again, and the photoelectric effect is the hinge that lets the same word run on a massless photon (all-at-once) or a massive electron braided through a machine's wiring. As literal braids the minimal hardware is tiny: a single qutrit needs three junctions ($B_3 \twoheadrightarrow \mathrm{PSL}(2,\mathbb{Z}) \to \mathrm{SL}(2,3) = 2T$, the same $2T$ as the gauge-connection holonomy), and four symplectic-transvection junctions already generate the full $\mathrm{Sp}(4,3)$. Their interaction graph (an edge where two junctions braid, $\langle v_i, v_j\rangle = \pm 1$; a non-edge where they commute) is a *path* $P_4$, so the minimal hardware is four junctions *in a line* — consecutive braid, non-consecutive commute. To run the machine on a given fabric is then to place that path: any wiring containing a simple four-junction path hosts it, which is every grid, ring, mesh, hypercube and almost every random fabric. Topology does genuinely matter in the pathological case — a pure star has no four-path, and the symplectic geometry forbids reshaping the interaction graph into a three-leaf star (mutually commuting junctions are isotropic, at most a Lagrangian) — but every real fabric qualifies, so "runs on any classical machine" becomes a concrete graph search for a four-junction path.

*(ii) The carrier as its own self-contained shell.* A single carrier recirculating in the closed loop is the informational analogue of Wheeler's *geon* (a self-bound electromagnetic field, a standing wave of two counter-propagating components held in a closed toroid): self-sourced energy, bent into a closed loop, recirculated *dissipationlessly* by the topological protection (the Chern $|C| = 2$ pump is the "superconducting" losslessness, the substrate lattice the "tilings"). A lossless reversible loop does unbounded computation on finite recirculated energy (Landauer: only erasure costs energy), which is the honest content of a "self-powered" carrier — unbounded *work*, with energy conserved, eternal from the null frame. (A single optical photon is $\sim 56$ orders from a literal *gravitational* geon; that forms at the Planck scale, where the substrate's derived Einstein equations govern.)

*(iii) The network is a fractal of nested shells.* Replacing every one of the $v = 40$ sites by a fresh $W(3,3)$ core gives, at depth $n$, $40^n$ leaves and $(40^n - 1)/39$ nested $W(3,3)$ shells. Reading each shell as a self-contained lossless, code-protected computer makes the network a self-similar tower in which each layer is the conceptual "Dyson sphere" of the shell outside it: every shell is an error-correcting code on its 40 sites, the outer code holographically containing the inner layers' logical information (boundary encodes bulk), and reversible shells composing into a single lossless network computer. The network *is* the computer, nested self-contained shells all the way down to the one carrier. (Witnesses: `analysis/w33_two_switch_generation.py`, `analysis/w33_photon_geon_dyson_sphere.py`, `analysis/w33_fractal_nested_dyson_spheres.py`. Honest scope: the "Dyson sphere" is informational/holographic — self-containment, losslessness, boundary-encodes-bulk — not a literal energy-harvesting megastructure, and no step creates energy.)

### B.28 Operator-on-photon self-applied circuit packet

One Choi leg is interpreted as the photonic mode state and the other Choi leg as the encoded optical action. The finite circuit is therefore modeled as a relation on one photon's own registers.

The self-applied model represents $I$, $X$, $Z$, $F_3$, and $S$ as encoded actions on the operator leg and checks them by trace-Choi overlap against the state leg.

The OAM operator witness distinguishes passive mode labels from active internal operator behavior by comparing label-only controls, operator-leg activation, basis covariance, and self-vs-external control tests.

### B.29 Internal optic algebra, physical dictionary, and falsifier regimes

The internal operations $I$, $X$, $Z$, $F_3$, and $S$ generate the finite single-qutrit projective Clifford action of size $216 = 9 \cdot 24$ on the state/operator Choi legs.

The operations are mapped to optical analogues: reference/no-op, OAM shift or prism step, spiral phase plate, tritter or mode mixer, and lens or phase-curvature analogue. The lens row remains calibration-level.

The falsifier simulator separates passive OAM labels from active internal-operator behavior. Only the internal-operator scenario passes the symbolic criteria: trace-Choi pattern, low leakage, basis covariance, and operator activation.

### B.30 Internal Clifford census, lens calibration, and passive-active protocol

The internal operations $I$, $X$, $Z$, $F_3$, and $S$ generate a 216-element single-qutrit projective Clifford action on the state/operator Choi legs. All 216 elements preserve the state/operator split in the internal-register model; only the identity stabilizes the identity Choi state itself. The 24 zero-translation frame changes preserve centered OAM opposition, while the 192 translated elements move the OAM origin and require recentering.

The lens calibration row is exact at the finite phase-signature level. On the centered OAM basis $\ell = -1, 0, +1$ with qutrit labels $2, 0, 1$, qutrit $S$ has $\omega$-exponent signature $[1, 0, 1]$, matching the centered quadratic lens phase $\omega^{\ell^2}$.

The passive-vs-active protocol uses centered OAM preparation, axial time-bin support, operator off/on controls, $I/X/Z/F_3/S$ settings, leakage checks, basis covariance, and external-reference comparison. It is a protocol design, not experimental evidence.

### B.31 OAM recentering, exact optical calibration, protocol table, and literature bridge

The 192 translated Clifford elements split into eight nonzero shift classes, each with 24 frame choices. The recentering correction is the inverse shift, with OAM-only, phase-only, and mixed shift classes.

The $S$ and $F_3$ calibrations are placed on the same centered OAM basis. The $S$ row has phase signature $[1, 0, 1]$, and the $F_3$ row is the finite three-mode mixer on the same qutrit labels.

The passive-vs-active protocol is converted into a paper-ready table covering preparation, passive controls, active gate settings, leakage, covariance, reference comparison, expected readouts, thresholds, failure modes, and claim tiers.

The OAM literature bridge supports high-dimensional qudit and same-photon register directions, but requires radial leakage and mode-overlap witnesses because radial and azimuthal spatial structure are coupled in realistic modes.

### B.32 Recentered witnesses, leakage regimes, and appendix splicer

The calibrated $I, X, Z, F_3, S$ witness applies directly to the centered frame class. Translated Clifford elements require inverse recentering before the same witness table is applied.

The radial leakage simulator turns symbolic shell leakage into pass/fail regimes, separating good core-gate behavior from radial leakage, visibility mismatch, and basis-covariance failures.

The operator/OAM appendix splicer follows the dry-run and idempotent splicer pattern. It records the appendix insert packets and changes the paper only when apply mode is used in checkout.

### B.33 Appendix validation, recentered protocol table, and claim ledger

The appendix packet is validated by a dry-run artifact checker: all insert paths exist, the bounded splice block is idempotent, and the target paper admits the expected splice anchor or bounded fallback.

The protocol table is expanded by recentering class. The resulting table has four Clifford recentering classes times five calibrated witnesses, with correction operators, leakage thresholds, failure modes, and claim tiers.

The claim ledger separates exact finite claims from calibration-level optics, engineering leakage, protocol witnesses, literature motivation, and blocked physical overclaims.

### B.34 Full operator/OAM appendix splice and recentered transaction ABI

The operator/OAM appendix is now part of the main Holonet paper through a bounded idempotent splice. The splice carries the operator-on-photon circuit model, the internal Clifford falsifier, the recentering protocol, the leakage and claim ledger, and the present synthesis block.

The architectural closure is the recenter-to-transaction ABI:

$$216 = 9 \cdot 24 = (1 + 2 + 2 + 4) \cdot 24.$$

The factor 24 is the centered BT1495 transaction-word kernel. The nine affine shifts are one centered class, two OAM-only shifts, two phase-only shifts, and four mixed shifts. Thus the class profile is

$$24, \quad 48, \quad 48, \quad 96,$$

and every translated internal Clifford action is executed by inverse recentering followed by one of the same 24 centered 72-tick transaction words.

One operation sweep therefore has $216 \cdot 72 = 15552$ ticks. The full five-witness $I, X, Z, F_3, S$ protocol has 77760 ticks. This is an exact finite ABI statement: it certifies the centered word reuse and the recentering burden. It does not claim measured OAM leakage, optical loss, or a finished calibrated device.

Recent OAM and high-dimensional time-bin experiments support the physical direction, including same-photon polarization/OAM gates, robust time-bin qudit control, directly measured topology in OAM-entangled states, and OAM-multiplexed diffractive mode processing [29, 30, 31, 32]. BT1588 keeps those sources in the literature-motivation tier only: they motivate the mode interface, multiplexing optics, and leakage tests, while the exact theorem remains the local finite $216 = 9 \cdot 24$ recenter ABI.

### B.35 LG radial covariance, full witness lane sheet, and OAM front end

The recenter ABI now has a radial-shell falsifier. Model the first three Laguerre–Gaussian radial shells by the stochastic channel family $L(\eta)$: diagonal entries $1 - \eta$, off-diagonal entries $\eta/2$. The BT1577 operation envelopes $I, Z, S, X, F_3$ all lie in this same channel family, so the radial recenter correction and the centered witness gate commute at shell level. For two envelopes,

$$\eta(a, b) = a + b - \frac{3}{2}ab.$$

Thus the $216 = 9 \cdot 24$ affine actions do not require 216 separate radial calibrations. They require one centered witness table plus the recenter tax. The symbolic worst case is a mixed OAM/phase recenter followed by $F_3$,

$$\eta_{\max} = 0.18296 < 0.20,$$

while the centered gate threshold remains the BT1577 value 0.10. This is a covariance simulator and falsifier, not measured beam leakage.

The complete witness replay is now compiled as a physical lane sheet:

$$5 \text{ gates} \times 9 \text{ recenter sectors} \times 24 \text{ centered words} \times 72 \text{ ticks} = 77760.$$

It is stored as 1080 exact 72-tick segments with the global tick formula

$$((((g \cdot 9) + s) \cdot 24 + w) \cdot 72 + t).$$

Every Hesse lane appears 12960 times, every detector slot appears 19440 times, native $D_4$ square-pulse words occupy 25920 ticks, and the remaining $S_4$ analyzer-relabel words occupy 51840 ticks.

The hardware interpretation is deliberately sharper. OAM-multiplexed diffractive optics are a proposed *front-end sectorizer*: a passive mode-processing layer can sort or fan out the nine affine recenter sectors before the exact 24-word transaction kernel runs. Recent OAM-MDNN work supports this as an optical mode-processing direction, but it does not prove a quantum gate, preserve coherence by itself, or calibrate loss. The promoted theorem remains local:

$$\boxed{9 \text{ recenter sectors} \times 24 \text{ centered words} = 216 \text{ exact finite ABI addresses,}}$$

with acceptance delegated to radial tomography, sector crosstalk measurements, and the 77760-tick witness replay.

## B.36 Lab tomography, LG alphabet, and the Hesse/T witness loop

The OAM front end is now written in a lab-facing form. BT1592 promotes three acceptance surfaces: a $9 \times 9$ sector confusion matrix, four radial-shell tomography rows, and six exact lane-replay checks. The synthetic sector fixture uses diagonal probability 0.93, off-diagonal probability 0.00875, and acceptance thresholds

$$p_{\text{diag}} \geq 0.90, \qquad p_{\text{off}} \leq 0.02.$$

The generated CSV ingest file has 91 rows, all passing, and is deliberately replaceable by bench data. This is a falsification harness, not measured OAM crosstalk.

BT1593 chooses the explicit Laguerre–Gaussian sector alphabet:

$$s = 3x + z, \qquad \ell = s - 4, \qquad p = x + 2z \pmod 3.$$

Thus the nine recenter sectors occupy symmetric OAM charges

$$\ell = -4, -3, -2, -1, 0, 1, 2, 3, 4,$$

with three modes in each radial shell $p = 0, 1, 2$. The 24-word centered selector is a separate handoff layer:

$$\text{address} = 24s + w, \qquad 0 \leq s < 9, \quad 0 \leq w < 24.$$

The architecture therefore uses 9 physical sector modes plus a 24-word selector, not 216 separate OAM channels. The 216 exact finite addresses split into 72 native $D_4$ square-pulse addresses and 144 $S_4$ analyzer-relabel addresses.

The universality port now lands in the same witness loop. BT1404 already made a Hesse/T microframe with

$$9 \text{ Hesse outcomes} \times 8 \text{ ticks} = 72 \text{ ticks}.$$

BT1590 already made each OAM witness segment 72 ticks. BT1594 overlays these rather than appending a second schedule:

$$1080 \text{ segments} \times 72 \text{ ticks} = 1080 \text{ Hesse/T microframes} = 77760 \text{ ticks}.$$

Each Hesse outcome appears once per segment, hence 1080 times total. The T-frame bit profile is $0 : 5400$ and $1 : 4320$, matching the parity rule

$$t = (r + p) \bmod 2.$$

This closes the finite architecture loop: the Clifford/OAM witness, the sector-mode front end, the tomography acceptance harness, and the explicit Hesse/T non-Clifford port now inhabit one replayable 77760-tick object. The claim boundary remains unchanged: this is ABI and schedule compatibility, not a physical Hesse-SIC device, injection-fidelity measurement, or fault-tolerant magic-state-yield theorem.

## B.37 The Witting reject shell as the universal fuel rail

The deepest closure in the current architecture is that the newest OAM/Hesse witness loop is not merely a validation sweep. It is the Witting matter shell itself. In the Witting delayed-query desk, each selected ray has

$$1 \text{ same} + 12 \text{ compatible} + 27 \text{ incompatible} = 40$$

possible queried rays. Therefore the incompatible ordered-pair shell has

$$40 \cdot 27 = 1080$$

records. BT1595 constructs an explicit bijection

$$40 \cdot 27 = 5 \cdot 9 \cdot 24$$

between those rejected Witting pairs and the BT1590/BT1594 witness segments. The per-gate refinement is even sharper:

$$9 \cdot 24 = 216 = 8 \cdot 27.$$

Thus each of the five witness gates consumes the full incompatible-target shell of eight Witting source rays. Each rejected Witting pair is one 72-tick contextual fuel segment, and each such segment carries the Hesse/T overlay.

BT1596 writes the resulting runtime economy:

$$520 \cdot 72 = 37440 \qquad \text{accepted communication/control ticks,}$$

$$1080 \cdot 72 = 77760 \qquad \text{contextual fuel ticks,}$$

and

$$1600 \cdot 72 = 115200 \qquad \text{ticks for the complete Witting ordered-pair cycle.}$$

The ratio of accepted control to contextual fuel is $13 : 27$, matching the Witting split $13/40$ versus $27/40$.

BT1597 packages this as a single universal transaction object. Accepted Witting pairs form the communication and control rail. Rejected Witting pairs are not discarded; they are exactly the OAM/Hesse contextual-fuel rail:

$$1080 = 5 \text{ gates} \cdot 9 \text{ OAM sectors} \cdot 24 \text{ words} = 40 \text{ rays} \cdot 27 \text{ incompatible targets}.$$

The Hesse/T microframe on that fuel rail supplies the explicit non-Clifford boundary required by the universal-computation contract. The claim remains a finite ABI and timing theorem: it does not assert optical coherence, loss tolerance, cryptographic security, or a fault-tolerant threshold.

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